A Major Project Report
on

RetinaPredict

by

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Certificate

This is to certify that

Rajveer Rai
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Have satisfactory completed this project entitled

RetinaPredict

Towards the fulfilment of the
BACHELOR OF ENGINEERING IN
COMPUTER ENGINEERING
as laid by University of Mumbai.

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Project Report Approval for B.E.

This project report entitled ***RetinaPredict*** by ***Rajveer Rai, Jeetkumar Rathod, Harshal Satam, Divyesh Shinde*** is approved for partial fulfilment of the degree of Computer Engineering.

Examiners

1. _____

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Declaration

We wish to state that the work embodied in this project titled "**RetinaPredict**" forms our own contribution to the work carried out under the guidance of "**Prof. Naina Kaushik**" at the Rajiv Gandhi Institute of Technology.

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Diabetic retinopathy (DR) represents one of the most consequential and rapidly escalating complications of diabetes mellitus, constituting a leading cause of preventable vision impairment and blindness among working-age adults globally. The insidious nature of this condition—characterized by an asymptomatic early course—frequently results in delayed clinical presentation. By the time patients manifest symptoms such as visual blurring, floaters, or scotomas, substantial retinal damage has already ensued, considerably diminishing the prospects for successful intervention and visual rehabilitation. Given its silent progression and increasing prevalence, timely detection of DR is paramount to preventing irreversible blindness and optimizing long-term visual prognosis in the diabetic population.

To address this pressing clinical challenge, this paper proposes an advanced deep learning-based framework for the automated detection and grading of diabetic retinopathy utilizing Convolutional Neural Networks (CNNs). CNNs demonstrate superior pattern recognition capabilities, rendering them particularly well-suited for the analysis of complex medical imagery. Through training on an extensive and diverse dataset of annotated retinal fundus images, the proposed model acquires the ability to identify cardinal pathological features—including microaneurysms, intraretinal hemorrhages, hard exudates, and cotton-wool spots—which constitute the established hallmarks of DR. Beyond mere detection, the system classifies disease severity according to a standardized grading scale, spanning mild non-proliferative diabetic retinopathy (NPDR) through proliferative diabetic retinopathy (PDR) and diabetic macular edema (DME), the most visually threatening manifestation.

Conventional DR screening paradigms depend heavily on the subjective interpretation of retinal images by ophthalmology specialists—a process that is inherently time-intensive, resource-dependent, and constrained by the limited availability of qualified personnel, particularly in rural and underserved healthcare settings. The proposed AI-driven diagnostic solution addresses these systemic limitations by delivering rapid, objective, and scalable image analysis capable of seamless integration into primary care and telemedicine infrastructures. This enables broad, cost-effective population screening and facilitates early therapeutic intervention at a stage when treatment yields the greatest clinical benefit. Ultimately, the incorporation of this deep learning-based diagnostic tool into contemporary healthcare delivery has the potential to fundamentally transform diabetic eye care by expanding screening coverage, enabling timely case identification, and alleviating the global burden of diabetes-related blindness.

Keywords: Diabetic Retinopathy, Deep Learning, Convolutional Neural Networks, Retinal Fundus Imaging, Early Detection, Medical Image Analysis, Automated Diagnosis, Severity Classification, Ophthalmology.

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List of Symbols

Symbol	Description
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
DR	Diabetic Retinopathy
NPDR	Non-Proliferative Diabetic Retinopathy
PDR	Proliferative Diabetic Retinopathy
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
CLAHE	Contrast Limited Adaptive Histogram Equalization
Grad-CAM	Gradient-weighted Class Activation Mapping
F1	F1-Score (Harmonic Mean of Precision and Recall)

Chapter 1

Introduction

1.1 Introduction Description

Diabetic Retinopathy (DR) is a severe and progressively debilitating microvascular complication of diabetes mellitus, representing one of the foremost causes of preventable blindness among the working-age population worldwide. The global burden of this condition continues to escalate in parallel with the rising prevalence of diabetes, creating an urgent and growing public health crisis. What makes DR particularly challenging from a clinical standpoint is its characteristically silent progression during the early and most therapeutically amenable stages of the disease. Patients frequently remain entirely asymptomatic until the condition has advanced to a stage at which irreversible retinal damage has already occurred, rendering timely intervention considerably more difficult and less effective. Pathological changes such as microaneurysms, intraretinal hemorrhages, hard exudates, and neovascularization develop gradually within the retina, often escaping the patient's awareness until vision loss becomes perceptible. This asymptomatic onset, combined with the global shortage of trained ophthalmologists—particularly in low- and middle-income countries—creates a critical diagnostic gap that traditional screening infrastructure is ill-equipped to address.

Conventional DR screening relies on periodic retinal examinations conducted by ophthalmologists or trained graders, a process that, while clinically effective, is inherently constrained by its dependency on specialist availability, manual interpretation, and significant time investment per patient. These limitations render large-scale population screening logistically challenging, particularly in rural and resource-limited healthcare settings where the burden of diabetes is often highest. The integration of artificial intelligence into ophthalmic diagnostics presents a compelling opportunity to overcome these systemic barriers. Deep learning, and more specifically Convolutional Neural Networks (CNNs), have demonstrated remarkable capability in the domain of medical image analysis, consistently achieving diagnostic performance comparable to—and in some cases surpassing—that of human experts. By automatically learning hierarchical representations of visual features directly from raw image data, CNNs are uniquely suited to detect the subtle morphological changes in retinal fundus images that are indicative of early-stage DR. To address this limitation, this project integrates Long Short-Term Memory (LSTM) networks with CNN-based feature extraction, forming a robust hybrid architecture specifically designed to enhance the depth and quality of DR classification.

The proposed CNN-LSTM hybrid model operates through a two-stage computational pipeline. In the first stage, the CNN component serves as a powerful spatial feature ex-

tractor, systematically identifying and encoding low- and high-level visual patterns from retinal fundus images—including lesion morphology, vascular abnormalities, and textural irregularities characteristic of progressive DR. In the second stage, the extracted feature sequences are passed into the LSTM network, which leverages its inherent capacity for sequential dependency modeling to capture temporal and contextual relationships among the extracted spatial features. This integration enables the model to move beyond pixel-level pattern recognition toward a more holistic, context-aware interpretation of retinal pathology, resulting in improved discriminative power across all severity grades of DR.

This report presents the complete design, implementation, and empirical evaluation of the proposed CNN-LSTM framework applied to the automated detection and severity grading of diabetic retinopathy. The system is trained and validated on a large, annotated dataset of retinal fundus images, encompassing the full spectrum of DR severity—from no DR through mild, moderate, and severe non-proliferative DR, to proliferative DR. Performance is rigorously assessed using quantitative metrics including classification accuracy, sensitivity, specificity, precision, F1-score, and area under the receiver operating characteristic curve (AUC-ROC), enabling a comprehensive and objective evaluation of the model's diagnostic reliability.

1.2 Organization of Report

A brief chapter-by-chapter overview is presented here.

- **Ch.1 Introduction:** This chapter provides an overview of the project, including the background, motivation, problem statement, objectives, scope, and significance of the proposed system. It sets the foundation by highlighting the challenges in the healthcare sector and the need for a digital solution like RetinaPredict.

- **Ch.2 Review of Literature:** This chapter reviews existing research, tools, and technologies related to automated DR detection systems. It presents a comparative analysis of previous works, identifies gaps in current systems, and justifies the novelty of the proposed approach.

- **Ch.3 Proposed System:** This chapter details the design and methodology of the RetinaPredict platform. It explains the system architecture, key modules, algorithms, data flow, and technologies used (Image Processing, Deep Learning, etc.). The implementation strategies and functional scope are also described.

- **Ch.4 Experimental Setup:** Explains the datasets, tools, and technologies used for model training and testing. It also outlines evaluation parameters, system configuration, and performance metrics.

- **Ch.5 Results & Discussions:** Details the step-by-step implementation results, workflow, and deployment approach. It also discusses validation methods and performance outcomes.

Chapter 2

Literature Review

Extensive research has focused on developing automated systems for Diabetic Retinopathy (DR) detection to support ophthalmologists and improve patient outcomes. Several deep learning and machine learning-based models have been proposed to address the diagnostic challenges associated with DR; however, their performance remains suboptimal in key areas. Critical challenges include achieving consistent sensitivity across all DR severity stages, ensuring generalizability to diverse and heterogeneous datasets, and managing the substantial computational demands of complex architectures. Consequently, designing robust, efficient, and highly accurate deep learning models for DR detection remains an active and clinically significant research area.

2.1 Survey of Existing Systems

A foundational study by Wang et al. employed multiple CNN architectures—including VGG16, AlexNet, and InceptionV3—trained using transfer learning for multi-stage DR classification. Among the architectures evaluated, InceptionV3 demonstrated the highest classification accuracy of 63.23%, attributed to its multi-scale feature extraction capabilities. While the study provided a valuable comparative analysis of CNN performance in medical imaging, it was constrained by a small and imbalanced dataset, limiting the model's ability to generalize across diverse patient populations and image acquisition conditions [1].

Mobeen-ur-Rehman et al. proposed a customized CNN architecture developed from scratch, specifically tailored to capture DR-specific retinal features. The model achieved a notably high classification accuracy of 98.15%, demonstrating strong performance in identifying localized retinal abnormalities such as microaneurysms, hemorrhages, and exudates. However, the study was conducted on a limited dataset without external validation, raising concerns regarding real-world generalizability and deployment feasibility in clinical screening environments [2].

Zhang et al. introduced an ensemble deep learning framework integrating adapted versions of ResNet50, InceptionV3, and DenseNet for automated DR identification and grading. By leveraging the complementary feature extraction strengths of each architecture, the system achieved a classification accuracy of 96.5%, surpassing individual CNN-based models in both robustness and lesion detection granularity. Despite these strengths, the reliance on a private dataset limited reproducibility, and the ensemble's elevated computational cost reduced its suitability for resource-constrained clinical deployment [3].

Shanthi and Sabeenian presented a modified AlexNet architecture for DR image classification, incorporating fine-tuned convolutional layers, optimized hyperparameters, and advanced preprocessing techniques. The model achieved an accuracy of 96.35%, demonstrating reduced overfitting and faster convergence compared to the standard AlexNet baseline. Nonetheless, the model was restricted to single-lesion detection per image and lacked validation on external datasets, limiting its applicability to multi-lesion clinical scenarios and broader patient populations [4].

Khan et al. proposed a hybrid deep learning model designated VGG-NIN, combining the hierarchical feature extraction depth of VGG with the nonlinear micro-network transformations of Network-in-Network (NIN) architecture. This hybrid approach demonstrated superior classification accuracy and robustness over single-architecture CNNs, with improved generalization across DR severity levels. However, the model exhibited considerable computational complexity, requiring substantial GPU resources and extended training durations, while its lack of large-scale cross-dataset validation continued to limit its clinical scalability [5].

Collectively, the reviewed studies demonstrate the significant progress achieved through CNN-based approaches in automating DR detection and grading. However, a consistent pattern of limitations emerges across the literature: restricted dataset sizes, insufficient external validation, high computational overhead, and the absence of contextual or sequential feature modeling that could further enhance classification performance. These gaps underscore the motivation for the present work, which proposes a hybrid CNN-LSTM architecture designed to address these deficiencies by integrating spatial feature extraction with sequential dependency modeling.

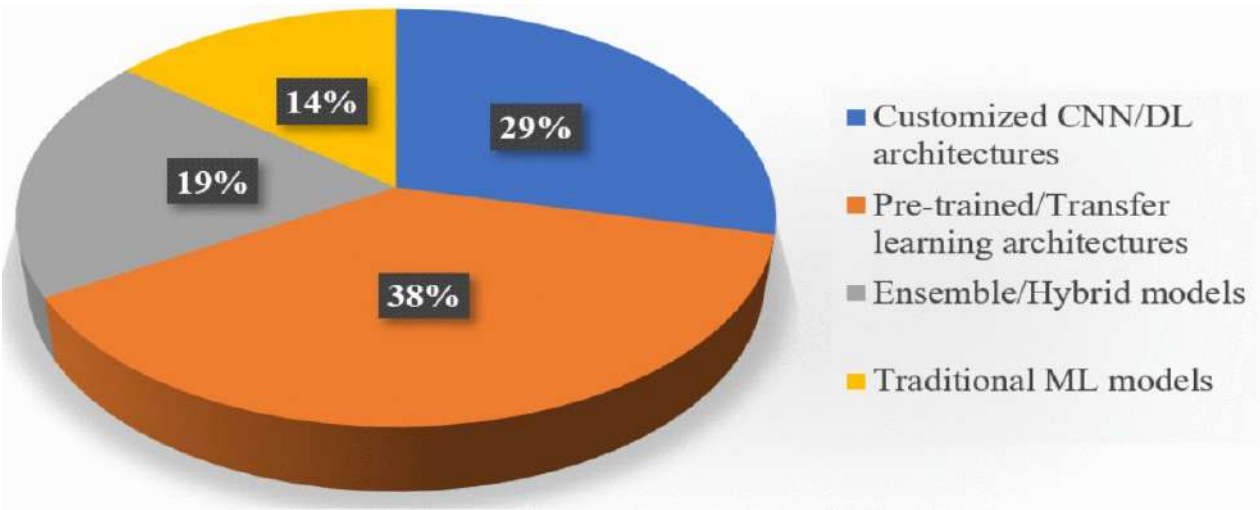


Figure 2.1: Distribution of ML/DL Model Architectures in Diabetic Retinopathy (DR) Detection

As depicted in Figure 2.1, proportional use of four major categories of machine learning and deep learning architectures in published research on automated DR detection is shown. Below is a detailed academic elaboration of each category:

Pre-trained / Transfer Learning Architectures (38%) — The dominant approach. Models such as VGG16, ResNet50, InceptionV3, EfficientNet, and DenseNet, originally trained on large datasets like ImageNet, are fine-tuned on retinal fundus image datasets.

Transfer learning addresses the perennial challenge of limited annotated medical data, enabling high sensitivity and specificity even with smaller DR-specific corpora.

Customized CNN / DL Architectures (29%) — The second-largest share. Researchers design task-specific convolutional pipelines tailored to the morphological characteristics of DR lesions—microaneurysms, hemorrhages, hard exudates, and neovascularization. These bespoke architectures often incorporate attention mechanisms, multi-scale feature extraction, or lesion-specific segmentation heads to optimize detection at various severity grades.

Ensemble / Hybrid Models (19%) — Combining multiple base learners or fusing CNN features with other modalities to improve robustness and reduce prediction variance. Hybrid approaches may also integrate traditional handcrafted features with deep embeddings, leveraging the complementary strengths of both paradigms.

Traditional ML Models (14%) — The smallest but historically significant category, encompassing SVMs, Random Forests, and k-NN classifiers operating on handcrafted features (GLCM, LBP, color histograms). While largely superseded by deep learning, these methods remain relevant in low-resource settings or as interpretable baselines.

2.2 Limitations of Existing Systems / Research Gap

1. Wang et al. [1] achieved a modest peak accuracy of only 63.23% using InceptionV3, indicating that standard CNN architectures, even when fine-tuned via transfer learning, remain insufficiently precise for reliable clinical DR screening.

2. The study by Wang et al. [1] was constrained by a small and class-imbalanced dataset, significantly restricting the model's capacity to generalise across heterogeneous patient populations and diverse retinal imaging conditions.

3. Mobeen-ur-Rehman et al. [2] trained and evaluated their customised CNN exclusively on a limited, intra-dataset setting without any external validation, raising serious concerns about the model's real-world clinical applicability.

4. The customised CNN proposed by Mobeen-ur-Rehman et al. [2] did not address computational efficiency or real-time deployment feasibility, both of which are critical requirements for large-scale clinical screening systems.

5. Zhang et al. [3] relied entirely on a private dataset for model evaluation, preventing reproducibility and making it impossible to benchmark performance against widely accepted public datasets such as EyePACS or Messidor.

6. The ensemble framework introduced by Zhang et al. [3], while achieving high accuracy, incurred substantially elevated computational costs, rendering the architecture unsuitable for large-scale or low-resource clinical deployment scenarios.

7. Shanthi and Sabeenian [4] demonstrated that their modified AlexNet architecture was capable of detecting only a single lesion type per retinal image, severely limiting its utility in multi-lesion classification tasks that are essential for comprehensive DR grading.

8. The modified AlexNet model proposed by Shanthi and Sabeenian [4] was not validated on external datasets, and neither scalability nor real-time inference performance was evaluated.

9. Khan et al. [5] reported that the VGG-NIN hybrid architecture exhibited considerable computational complexity, necessitating substantial GPU resources and prolonged training durations.

10. Across all reviewed studies, a consistent and critical limitation is the absence of sequential or contextual feature modelling. Each architecture restricts itself to spatial feature extraction alone, failing to capture the inter-regional dependencies and progressive pathological patterns across retinal structures.

Research Gap: A thorough review of the existing literature reveals a substantial and multifaceted research gap in the automated detection and grading of Diabetic Retinopathy. While the reviewed studies have each demonstrated meaningful progress through the application of CNN-based architectures, they collectively share a fundamental architectural limitation: an exclusive reliance on spatial feature extraction without any mechanism for modelling sequential dependencies or contextual relationships among extracted retinal features. No existing study has proposed a hybrid architecture that integrates CNN-based spatial feature extraction with Long Short-Term Memory (LSTM) networks for sequential dependency modelling—a combination uniquely positioned to address the dual challenge of feature richness and contextual classification accuracy. The proposed CNN-LSTM hybrid framework directly addresses these gaps by combining deep spatial feature learning with sequential contextual modelling, validated on a diverse dataset, offering a more comprehensive, clinically viable, and scalable solution for automated DR diagnosis.

2.3 Problem Statement and Objectives

Diabetic Retinopathy (DR) remains one of the most prevalent and clinically consequential complications of diabetes mellitus, constituting a leading cause of preventable vision impairment and blindness among the global working-age population. The disease's characteristically asymptomatic progression during its early and most therapeutically manageable stages presents a fundamental challenge to timely clinical intervention. The prevailing standard of care for DR diagnosis relies predominantly on the manual examination and interpretation of retinal fundus images by qualified ophthalmologists. While clinically sound, this approach is inherently constrained by several systemic limitations: it is time-consuming and labor-intensive, subject to inter-observer variability and human error, which can compromise diagnostic consistency and reliability.

Objectives:

1. To develop a deep learning-based system for the automated detection and classification of Diabetic Retinopathy from retinal fundus images.

2. To implement and systematically compare the performance of various established CNN architectures, including AlexNet, ResNet50, DenseNet121, and InceptionV3, for this task.

3. To design and evaluate a novel hybrid deep learning model combining CNNs (Inception-ResNetV2) and Long Short-Term Memory (LSTM) networks to improve classification accuracy.

4. To evaluate all models using key performance metrics such as accuracy, precision, sensitivity, and F1-score to identify the most effective architecture.

5. To create a system capable of accurately classifying DR into its five severity levels (No DR, Mild, Moderate, Severe, Proliferative) to assist clinicians in diagnosis and treatment planning.

2.4 Scope

The scope of this project encompasses the design, development, and evaluation of an intelligent, automated diagnostic system for the detection and severity grading of Diabetic Retinopathy (DR) using a hybrid deep learning architecture combining CNNs and LSTM networks. The system operates on retinal fundus images and is intended to serve as a clinically viable decision-support tool. The project trains and evaluates nine interchangeable deep learning architectures—ranging from standard CNNs such as AlexNet, ResNet50, InceptionV3, EfficientNetB0, and DenseNet121, to advanced hybrid models including CNN+LSTM, MobileNetV2+LSTM, and the primary InceptionResNetV2+LSTM.

The functional and technical scope includes:

- **Multi-class DR severity classification** across five standardized grades—No DR, Mild, Moderate, Severe, and Proliferative DR.

- **Hybrid CNN-LSTM architecture** as the primary model, leveraging CNN layers for spatial retinal feature extraction and LSTM layers for modelling sequential dependencies.

- **Explainable AI via Grad-CAM** heatmap generation, visually highlighting specific retinal regions that influence the model's diagnostic decision.

- **Automated medical PDF report generation** providing patients and clinicians with a structured, downloadable diagnostic summary.

- **Dual-role web application** built on Streamlit, separating the administrative model engineering workflow from the patient-facing clinical inference workflow.

- **Comprehensive evaluation framework** utilizing medically appropriate metrics including Quadratic Cohen's Kappa, Macro F1-Score, and One-vs-Rest AUC-ROC.

- **Image preprocessing pipeline** incorporating CLAHE-based contrast enhancement and pixel normalization.

- **Scalability and adaptability** of the modular architecture allowing new deep learning models to be integrated via the unified Model Factory.

Chapter 3

Proposed System

3.1 Proposed System Statement

The proposed system, RetinaPredict, is a comprehensive AI-driven ophthalmic diagnostic platform designed to provide automated, accurate, and clinically interpretable detection and severity grading of Diabetic Retinopathy from retinal fundus images. The system is motivated by the critical need to address the diagnostic bottleneck created by the global shortage of trained ophthalmologists, the asymptomatic progression of early-stage DR, and the consequent delays in treatment that lead to irreversible vision loss in diabetic patients. It integrates a hybrid deep learning architecture combining Convolutional Neural Networks (CNN) for hierarchical spatial retinal feature extraction and Long Short-Term Memory (LSTM) networks for sequential contextual dependency modelling, enabling fine-grained, multi-class classification across all five internationally recognized DR severity stages—No DR, Mild Non-Proliferative DR, Moderate Non-Proliferative DR, Severe Non-Proliferative DR, and Proliferative DR.

Beyond classification, the system incorporates a clinically robust preprocessing pipeline for retinal image enhancement, Gradient-weighted Class Activation Mapping (Grad-CAM) for explainable AI visualization that highlights pathologically relevant retinal regions, and an automated PDF diagnostic report generation module that produces patient-ready clinical summaries inclusive of the fundus image, Grad-CAM overlay, severity grade, and per-stage confidence scores. Through a secure, dual-role Streamlit interface separating patient-facing inference workflows from administrator-facing model engineering workflows, the system ensures that both clinical end users and technical operators can interact with the platform in a manner appropriate to their respective roles, enabling data-driven, transparent, accountable, and clinically responsible diabetic eye care.

3.2 Analysis

The proposed RetinaPredict system is designed as a unified, end-to-end diagnostic application that consolidates retinal image ingestion, multi-stage preprocessing, hybrid deep learning inference, explainability visualization, comprehensive performance evaluation, and structured clinical report generation within a single Python-based Streamlit environment. Unlike traditional distributed multi-tier web architectures that require separate frontend, backend API, and database services, the system deliberately adopts a modular single-process

application design optimized for research prototyping, clinical demonstration, and ease of deployment.

The system analysis commenced with a comprehensive stakeholder and requirements gathering process spanning three primary user roles. The **Patient** represents diabetic individuals requiring periodic retinal screening who interact with the system to receive automated DR severity assessments without direct specialist involvement. The **Ophthalmologist** represents the qualified clinical authority who reviews and validates the system's diagnostic outputs and issues formal treatment recommendations where warranted. The **System Admin** represents the technical or research professional responsible for ongoing operational management and continuous improvement of the deep learning pipeline.

From this stakeholder analysis, the system's functional requirements were decomposed into five principal modules: the Data Engineering and Preprocessing Module, the Model Training and Management Module, the Inference and Prediction Module, the Explainability and Visualization Module, and the Report Generation Module.

The application is organized across four logical internal components ensuring a clear separation of functional concerns. The **Interface Component** renders role-specific pages entirely through Streamlit's Python rendering engine. The **Application Logic Component** manages session-state-based role authentication, page routing, preprocessing pipeline orchestration, and report compilation workflows. The **Deep Learning Inference Component** encapsulates all nine interchangeable CNN and hybrid CNN-LSTM architectures within a unified Model Factory design pattern. The **Data Component** handles all dataset ingestion, multi-source consolidation, outlier exclusion, stratified train-validation-test splitting, dynamic class weight computation, and image augmentation pipeline management.

Data collection consolidates retinal fundus images from multiple independent source datasets into a single unified repository. Data augmentation is applied exclusively to the training split, with stochastic transformations including random horizontal and vertical flipping, rotation, zoom, shear, and brightness adjustment applied during each training epoch. The dataset is partitioned into independent train, validation, and test splits using stratified sampling that preserves the proportional class distribution of all five DR severity grades across every subset.

3.3 Design Details

3.3.1 System Interaction

The RetinaPredict is architected as a comprehensive, end-to-end deep learning application built upon a modular, single-tier design philosophy using the Streamlit framework. Unlike traditional multi-tier web systems, the application consolidates the user interface, application logic, and model inference pipeline within a unified Python environment, which simplifies deployment while maintaining clear internal separation of functional concerns through modular code organization.

The overall application is internally structured across four logical components: the Presentation Component, the Application Logic Component, the Deep Learning Inference Component, and the Data Component. Each component is independently organized to ensure maintainability, readability, and scalability. The Presentation Component is implemented using Streamlit, which renders the interactive web interface for both administrative and patient-facing workflows directly from Python, eliminating the need for a separate fron-

tend framework. The Application Logic Component manages session-state-based authentication, page routing, preprocessing orchestration, and report generation. The Deep Learning Inference Component contains interchangeable model architectures unified under a Model Factory pattern, enabling seamless switching between architectures without disrupting the broader application. The Data Component handles dataset ingestion, stratified splitting, class weighting, and image augmentation pipelines using NumPy, Pandas, and OpenCV.

It is important to note that the system does not employ a separate frontend framework, dedicated REST API layer, or standalone database server. All operations—from image upload and preprocessing through model inference and PDF report generation—are executed within the same Python process. The system follows a classic supervised machine learning pipeline—from raw retinal image ingestion through preprocessing, feature extraction, classification, explainability generation, and final report delivery—ensuring a reproducible, transparent, and clinically interpretable diagnostic workflow at every stage.

Dual-Role System Design: A defining design decision of this system is the implementation of a secure, session-state-managed dual-role interface that clearly separates two distinct operational contexts within the same Streamlit application. The **Admin Role**, intended for ophthalmologists, researchers, or system engineers, provides access to the full model engineering workflow within the application. This includes architecture selection, hyperparameter configuration, live epoch-by-epoch training visualization, post-training evaluation via confusion matrices and ROC curves, and side-by-side multi-model comparison. The **User Role**, intended for patients or clinical staff, provides a streamlined inference interface. Users upload a retinal fundus image through a Streamlit file uploader widget, which is automatically preprocessed and passed through the active deployed model.

Use Case Design:

The use case design of the system defines the complete set of interactions between external actors and the functional capabilities of the Streamlit application. As illustrated in Figure 3.1, the system identifies three primary actors—the Patient, the Ophthalmologist, and the System Admin—each engaging with a distinct and well-defined set of use cases within the system boundary.

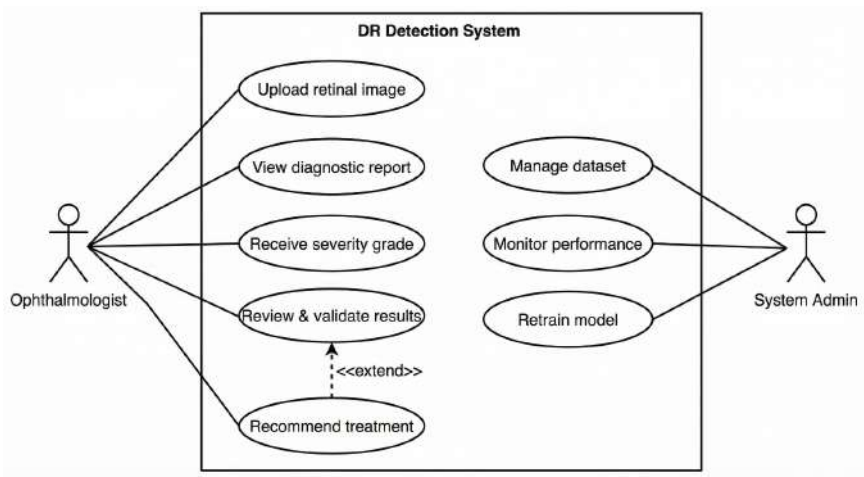


Figure 3.1: Use Case Diagram

Actors:

- **Ophthalmologist** – interacts with the system to upload images, view reports, receive severity grades, review results, and recommend treatment.
- **System Admin** – manages datasets, monitors system performance, and retrains the detection model.

Relationships:

- **Associations** – Actors are connected to their respective use cases: Ophthalmologist → Upload image, View report, Receive severity grade, Review & validate results, Recommend treatment. System Admin → Manage dataset, Monitor performance, Retrain model.
- **Extend** – *Recommend treatment* extends *Review & validate results*, indicating treatment is recommended only after validation.

3.3.2 System Architecture

The proposed system is architected as a comprehensive, end-to-end deep learning diagnostic application implemented entirely within a unified Python environment using Streamlit as the application framework. The architecture is designed to follow a structured, sequential data and model pipeline—from raw dataset ingestion through preprocessing, model training, and final DR prediction—ensuring reproducibility, clinical interpretability, and operational clarity at every stage. The overall pipeline can be divided into two primary phases: the Data Engineering and Model Training Phase, and the Inference and Prediction Phase.

Phase 1 — Data Engineering and Model Training Pipeline

Dataset Acquisition and Consolidation: The pipeline originates with the acquisition and consolidation of retinal fundus image data from multiple sources. The system accepts multiple independent datasets as input, which are merged into a single unified Combined dataset. During this consolidation step, outlier images—those exhibiting severe artifacts, poor focus, or insufficient retinal coverage—are identified and excluded to ensure data quality.

Preprocessing: Once consolidated, the combined dataset is passed through a dedicated preprocessing pipeline. Operations applied include:

- **CLAHE Enhancement** — Contrast Limited Adaptive Histogram Equalization is applied using OpenCV to amplify the local contrast of retinal images, enhancing the visibility of fine pathological structures.
- **Resizing and Normalization** — All images are resized to a uniform spatial resolution and pixel intensity values are normalized to the $[0, 1]$ range.
- **Noise Reduction and Quality Standardization** — Additional image quality corrections are applied to ensure consistent preprocessed output.

Dataset Splitting: The preprocessed data is partitioned into three independent subsets using stratified splitting:

- **Train Set** — The primary subset used for model weight optimization during training.

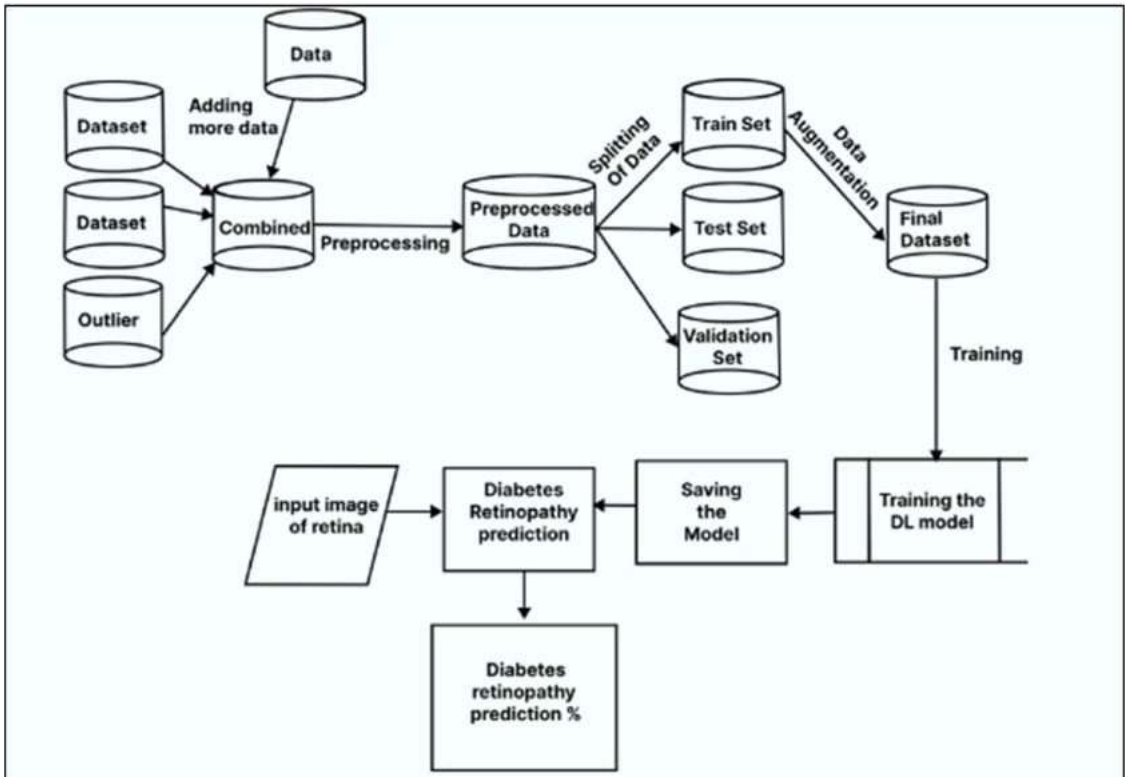


Figure 3.2: System Architecture

- **Validation Set** — An independent subset used during training to monitor generalization performance and implement early stopping.
- **Test Set** — A completely held-out subset used exclusively for final, unbiased performance evaluation.

Data Augmentation: To enhance the diversity of the training data, the Train Set undergoes systematic data augmentation including random horizontal and vertical flipping, rotation, zoom, shear, and brightness adjustment.

Model Training: The Final Dataset is passed into the deep learning training pipeline, where one of the available hybrid CNN-LSTM architectures is selected and trained using the TensorFlow 2/Keras framework. Dynamic class weighting is applied during training to compensate for class imbalance. Upon convergence, the trained model weights are serialized and saved to disk in HDF5 format as a `.h5` file.

Phase 2 — Inference and Prediction Pipeline

Input Image Acquisition: The inference phase is initiated when a patient or clinical user submits a raw retinal fundus image through the Streamlit user interface in standard image formats including `.png` and `.jpg`.

Preprocessing for Inference: Before the input image is passed to the model, the same preprocessing pipeline applied during training—contrast enhancement, resizing, and pixel normalization—is applied to the uploaded image in real time.

DR Prediction: The preprocessed image is passed to the saved CNN-LSTM model. The model performs a forward pass through the full InceptionResNetV2+LSTM architecture,

producing a five-class softmax probability vector representing the model's confidence for each DR severity grade.

Severity Grade Output: The final output is the DR severity prediction expressed as both a categorical grade label and a percentage confidence score. This output is displayed in the Streamlit interface with colour-coded severity indicators for immediate clinical interpretability.

Supporting Modules

The **Explainability Module** implements Grad-CAM to generate heatmap overlays on the input retinal image, visually identifying the specific retinal regions that were most influential in determining the model's severity prediction.

The **Report Generation Module** uses the ReportLab library to compile a structured, patient-ready PDF diagnostic report containing the input retinal image, predicted severity grade, confidence score, and Grad-CAM heatmap overlay.

The **Performance Monitoring Module**, accessible exclusively through the admin interface, provides a comprehensive evaluation dashboard presenting the confusion matrix, per-class ROC curves, Quadratic Cohen's Kappa, Macro F1-Score, and AUC metrics.

3.4 Methodology

3.4.1 CNN + LSTM

The CNN + LSTM model constitutes the foundational baseline hybrid architecture implemented within the RetinaPredict diagnostic system, serving the dual purpose of validating the core architectural hypothesis that integrating convolutional spatial feature extraction with recurrent sequential dependency modelling yields superior multi-class DR severity classification performance. Unlike the primary InceptionResNetV2+LSTM model which leverages the deep representational capacity of an ImageNet pre-trained backbone, this architecture employs a custom-designed lightweight CNN feature extractor constructed entirely from scratch and trained solely on the retinal fundus training corpus, coupled with an LSTM recurrent layer for contextual sequential modelling and a multi-layer fully connected classification head for final DR severity grade prediction.

Input Representation and Temporal Sequence Formulation: The model is designed to accept an input tensor structured as a temporal sequence, where the batch dimension is followed by a temporal dimension of value one representing the single retinal fundus image treated as a one-frame time series. The deliberate formulation of a static two-dimensional retinal image as a one-dimensional temporal sequence enables the downstream LSTM recurrent layer to process the CNN-extracted feature embedding within a formally defined sequential processing framework.

Global Average Pooling — Spatial Feature Compression: Following the TimeDistributed application of the custom CNN, a TimeDistributed Global Average Pooling operation is applied to spatially compress the three-dimensional output feature map tensor of the final convolutional block into a compact one-dimensional feature embedding vector for each temporal frame. Global Average Pooling computes the arithmetic mean of the activation values across the complete spatial extent of each individual feature map channel independently.

LSTM Layer — Sequential Contextual Dependency Modelling: The recurrent sequential modelling component is implemented as a Long Short-Term Memory network that

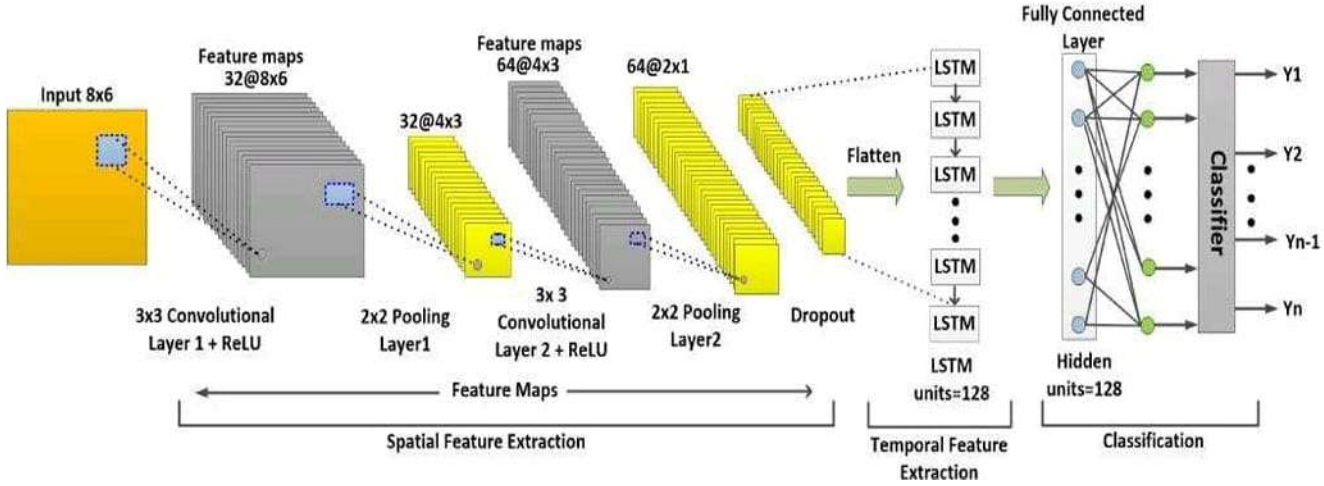


Figure 3.3: Architectural Overview of the CNN + LSTM Hybrid Model

processes the compact CNN-extracted feature embeddings as an ordered temporal sequence. The LSTM architecture addresses the principal limitation of standard Recurrent Neural Networks through the explicit introduction of a persistent learnable cell state vector and three multiplicative gating mechanisms—the forget gate, the input gate, and the output gate—that selectively regulate the flow, retention, and propagation of information through the recurrent network.

Table 3.1: CNN + LSTM Architecture Layer Summary

Layer	**Output Shape**	**Configuration**
Input	(N, 1, H, W, 3)	Single image as 1-frame temporal sequence
TD Conv Block 1	(N, 1, H/2, W/2, 32)	Conv2D + BatchNorm + ReLU + MaxPool(2×2)
TD Conv Block 2	(N, 1, H/4, W/4, 64)	Conv2D + BatchNorm + ReLU + MaxPool(2×2)
TD Conv Block 3	(N, 1, H/8, W/8, 128)	Conv2D + BatchNorm + ReLU + MaxPool(2×2)
TD GlobalAverage-Pooling2D	(N, 1, 128)	Spatial mean aggregation per channel
LSTM + BatchNorm	(N, 256)	Gated recurrent units, recurrent dropout
Dense FC1 + Dropout	(N, 128)	ReLU activation, dropout regularization
Output Dense	(N, 5)	Softmax five-class posterior distribution

3.4.2 InceptionResNetV2 + LSTM

The InceptionResNetV2 + LSTM model constitutes the primary and clinically deployed diagnostic architecture within the RetinaPredict system, representing the most sophisticated and highest-performing model among the three hybrid architectures implemented. The model is built upon the transfer learning paradigm, leveraging the deep representational capacity of the InceptionResNetV2 CNN pre-trained on the large-scale ImageNet dataset as a powerful spatial feature extraction backbone, integrated with a Long Short-Term Memory recurrent network for sequential contextual dependency modelling, and culminating in a fully connected multi-layer classification head that produces a five-dimensional softmax posterior probability distribution over the DR severity grades.

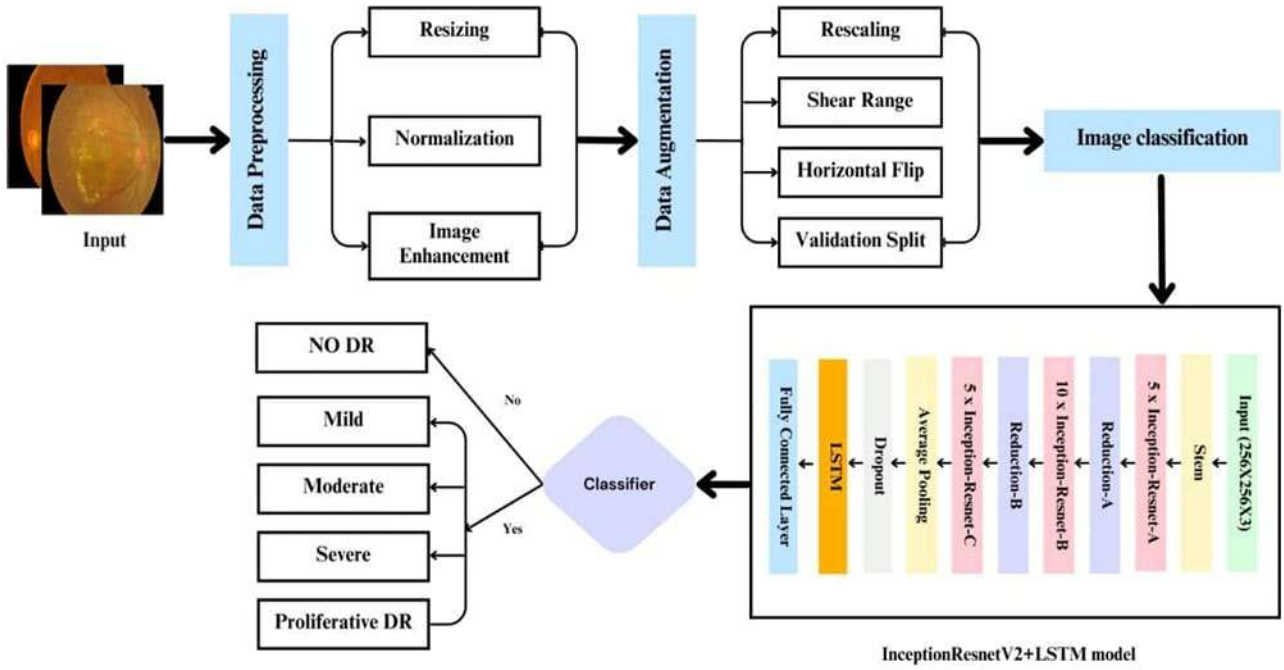


Figure 3.4: Architectural Overview of the InceptionResNetV2 + LSTM Hybrid Model

InceptionResNetV2: The spatial feature extraction component is built upon InceptionResNetV2, a deeply capable hybrid convolutional architecture that combines the multi-scale filter philosophy of the Inception family with the residual skip connection mechanism of ResNet. The Inception modules apply convolutional filters of multiple kernel sizes—including 1×1, 3×3, and 5×5—in parallel across the same input feature map, concatenating their outputs to produce a rich multi-scale feature representation. The InceptionResNetV2 backbone is initialized with weights pre-trained on the ImageNet dataset containing over 1.2 million images across 1,000 object categories, transferring rich, generalizable hierarchical visual feature representations to the retinal fundus image classification domain.

Global Average Pooling: Following TimeDistributed application of the InceptionResNetV2 backbone, a TimeDistributed Global Average Pooling operation compresses the three-dimensional output feature map tensor into a compact one-dimensional feature embedding vector of 1,536 dimensions for each temporal frame.

LSTM Layer: The recurrent sequential modelling component is implemented as an LSTM layer with 256 hidden units that processes the 1,536-dimensional CNN feature embedding through its gated computational mechanism. The LSTM is configured with `return_sequences=Fal` returning only the terminal consolidated hidden state output of 256 dimensions. A dropout rate of 0.3 is applied to the recurrent connections during training to prevent co-adaptation of recurrent units.

Classification Head: The classification head maps the LSTM's 256-dimensional consolidated contextual representation to the five-class DR severity output through a two-stage fully connected projection network. The first Dense layer of 512 neurons with ReLU activation is followed by Dropout at 0.4. The second Dense layer of 256 neurons with ReLU activation is followed by Dropout at 0.3. The terminal output Dense layer of 5 neurons applies the Softmax activation function.

Table 3.2: InceptionResNetV2 + LSTM Architecture Layer Summary

Layer	**Output Shape**	**Configuration**
Input	(N, 1, 224, 224, 3)	Single image as 1-frame temporal sequence
TD InceptionResNetV2	(N, 1, 5, 5, 1536)	ImageNet pre-trained, frozen weights
TD GlobalAveragePooling2D	(N, 1, 1536)	Spatial mean per channel, 1536-dim embedding
LSTM	(N, 256)	256 units, dropout 0.3, return final state
BatchNormalization	(N, 256)	Activation distribution normalization
Dense FC1 + Dropout	(N, 512)	ReLU activation, dropout rate 0.4
Dense FC2 + Dropout	(N, 256)	ReLU activation, dropout rate 0.3
Output Dense	(N, 5)	Softmax five-class posterior distribution

3.4.3 MobileNetV2 + LSTM

The MobileNetV2 + LSTM model constitutes the second hybrid architecture implemented within the RetinaPredict diagnostic system, positioned between the computationally minimal CNN+LSTM baseline and the high-capacity primary InceptionResNetV2+LSTM model. The model is designed around the principle of achieving a clinically meaningful balance between classification accuracy and computational efficiency, making it particularly well-suited for deployment scenarios where inference speed and memory constraints are as important as diagnostic precision.

CNN Backbone — MobileNetV2: The spatial feature extraction component is built upon MobileNetV2, a highly efficient lightweight convolutional neural network architecture specifically designed to minimize computational cost and memory footprint. The defining computational innovation of MobileNetV2 is the depthwise separable convolution, which factorizes the standard two-dimensional convolution into a depthwise convolution followed

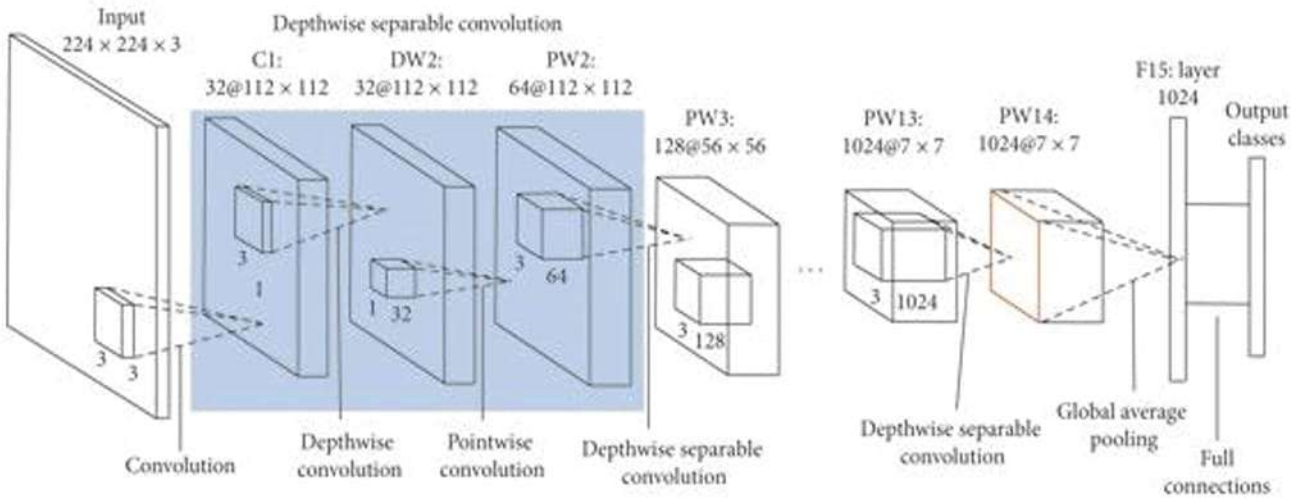


Figure 3.5: Architectural Overview of the MobileNetV2 + LSTM Hybrid Model

by a pointwise 1×1 convolution. This factorization dramatically reduces the total number of multiply-accumulate operations required per convolution. The MobileNetV2 backbone produces a 1,280-dimensional feature embedding after Global Average Pooling.

LSTM Layer: The recurrent sequential modelling component is implemented as an LSTM layer with 128 hidden units—deliberately configured with a smaller hidden state dimensionality than the 256-unit LSTM employed in the primary InceptionResNetV2+LSTM model, reflecting the architectural efficiency philosophy of this design.

Classification Head: The classification head consists of a single fully connected Dense layer of 256 neurons with ReLU activation followed by Dropout at 0.4, and the terminal output Dense layer of 5 neurons with Softmax activation.

Chapter 4

Experimental Setup

4.1 Details of Database

The dataset utilized in this study was sourced from Kaggle, one of the most widely recognized and extensively used open-access repositories for machine learning and medical imaging research. The dataset comprises 3,663 high-resolution colored retinal fundus images, each captured using standard fundus photography equipment under clinical imaging conditions. These images are pre-annotated and classified into five distinct categories corresponding to the internationally recognized severity scale of Diabetic Retinopathy:

- **No DR (Stage 0)** — No pathological changes detected in the retina.
- **Mild Non-Proliferative DR (Stage 1)** — Presence of at least one microaneurysm.
- **Moderate Non-Proliferative DR (Stage 2)** — More than just microaneurysms, but less than severe NPDR.
- **Severe Non-Proliferative DR (Stage 3)** — Extensive intraretinal hemorrhages and microvascular abnormalities.
- **Proliferative DR (Stage 4)** — New abnormal blood vessel formation (neovascularization).

The complete dataset of 3,663 images was partitioned into three independent, non-overlapping subsets:

- **Training Set** — 2,490 images (≈68% of dataset), subjected to full data augmentation transformations.
- **Test Set** — 439 images (≈12% of dataset), maintained as a completely held-out evaluation corpus.
- **Validation Set** — 733 images (≈20% of dataset), used during training to monitor generalization performance.

A significant characteristic of this dataset that substantially influences the model design and training strategy is the pronounced class imbalance across the five DR severity categories. The No DR class is heavily over-represented, with Stage 0 containing the largest

number of images (2,810), while Stages 3 and 4—Severe and Proliferative DR—contain considerably fewer samples (421 and 543 respectively). To address this, the system implements dynamic class weighting computed from the training set's class frequency distribution, and employs stratified splitting to ensure that each subset maintains a proportionally consistent representation of all five DR severity classes.

4.2 Performance Evaluation Parameters

5.1 Accuracy

Accuracy is the most fundamental classification metric, defined as the ratio of correctly classified samples to the total number of samples across all five DR severity classes. While accuracy provides an immediate high-level indication of overall model performance, it is considered an insufficient standalone metric for this system due to the significant class imbalance in the retinal dataset.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Figure 4.1: Accuracy Formula

5.2 Precision

Precision measures the proportion of images predicted as belonging to a particular DR severity class that were genuinely of that class, quantifying the model's exactness and its tendency to generate false positive diagnoses. High precision is particularly important in DR screening to avoid unnecessary clinical referrals.

$$Precision = \frac{TP}{TP + FP}$$

Figure 4.2: Precision Formula

5.3 Recall (Sensitivity)

Recall, also referred to as Sensitivity in the clinical context, measures the proportion of true positive DR cases that the model correctly identifies. Recall is arguably the most clinically critical metric in this system, as a false negative—failing to detect a genuine case of severe or proliferative DR—carries significantly higher clinical consequences than a false positive.

$$Recall = \frac{TP}{TP + FN}$$

Figure 4.3: Recall Formula

5.4 F1-Score

The F1-Score is the harmonic mean of Precision and Recall, providing a single balanced metric that accounts for both the model's exactness and its completeness simultaneously. It is particularly valuable in the context of imbalanced medical datasets.

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Figure 4.4: F1-Score Formula

5.5 Quadratic Cohen's Kappa

Quadratic Cohen's Kappa is the primary evaluation metric of this system and the gold standard for ordinal medical severity grading tasks. Unlike accuracy, Kappa measures the agreement between the model's predicted severity grades and the true grades while correcting for the agreement that would be expected by chance alone. The quadratic weighting penalizes predictions that are further from the true severity grade more severely than adjacent misclassifications.

5.6 AUC-ROC

The AUC-ROC metric quantifies the model's ability to discriminate between positive and negative cases for each DR severity class independently, using a One-vs-Rest multi-class strategy. For each class, the ROC curve plots the True Positive Rate (Recall) against the

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}$$

Figure 4.5: Quadratic Cohen's Kappa Formula

False Positive Rate across all possible classification thresholds, and the AUC summarizes this curve as a single scalar value between 0 and 1.

$$AUC = \int_0^1 TPR \, d(FPR)$$

Figure 4.6: AUC-ROC Formula

5.7 Confusion Matrix

The confusion matrix provides a complete and granular breakdown of the model's classification behavior across all five DR severity classes simultaneously, presenting the counts of true positive, false positive, true negative, and false negative predictions for every class combination in a structured matrix format.

5.8 Training History (Loss and Accuracy Curves)

Training history curves track the evolution of both training and validation accuracy and loss values across all training epochs, providing a dynamic view of model convergence behavior and generalization capability throughout the optimization process. The system renders these curves as live epoch-by-epoch charts within the Streamlit admin interface during training.

4.3 Software and Hardware Setup

4.3.1 Software Requirements

1. **Operating System:** Windows 10/11, Linux (Ubuntu 20.04+), or macOS—any platform supporting Python 3.9 or higher.

2. **Application Framework:** Streamlit—serves as the unified interface and application layer for both the admin engineering dashboard and the patient-facing inference workflow.

3. **Deep Learning Framework:** TensorFlow 2.x / Keras—used for model architecture definition, training, weight serialization, gradient computation for Grad-CAM, and inference.

4. **Data Engineering:** NumPy, Pandas—used for array manipulation, dataset management, class weight computation, and tabular metric handling.

5. **Model Evaluation:** Scikit-learn—utilized for stratified dataset splitting, confusion matrix generation, classification report computation, and ROC-AUC metric calculation.

6. **Visualization:** Matplotlib, Plotly, Seaborn—employed for training history charts, confusion matrix heatmaps, per-class ROC curves, and side-by-side model comparison charts.

7. **Report Generation:** ReportLab—used for compiling and exporting structured, patient-ready PDF diagnostic reports containing the fundus image, Grad-CAM overlay, severity grade, and confidence scores.

8. **Authentication and Session Management:** Streamlit Session State—enforces role-based access control separating admin and patient user contexts.

9. **Development Tools:** Python 3.9+, pip package manager, Visual Studio Code or PyCharm as the recommended IDE, Git for version control.

4.3.2 Hardware Requirements

1. **Processor:** Intel Core i5 / AMD Ryzen 5 (8th generation or higher) for development and inference; Intel Xeon or AMD EPYC recommended for server-side model training.

2. **RAM:** Minimum 8 GB for inference and application execution; 16 GB or higher strongly recommended for model training, particularly for large architectures such as InceptionResNetV2+LSTM.

3. **Storage:** Minimum 512 GB SSD for local development, model weight storage, and retinal image dataset management; 1 TB+ SSD recommended for server deployment.

4. **Internet Connection:** Stable broadband connection required for initial dependency installation via pip, Streamlit cloud deployment if applicable, and remote dataset access.

Chapter 5

Results & Discussions

The Diabetic Retinopathy Detection System successfully demonstrates the potential of hybrid deep learning in enabling early, automated, and accurate DR diagnosis from retinal fundus images. By combining CNN-based spatial feature extraction with LSTM-based sequential modelling, integrated within a unified Streamlit application alongside CLAHE preprocessing, Grad-CAM explainability, and automated PDF report generation, the system delivers clinically interpretable and actionable diagnostic outputs across all five DR severity grades. The use of medically appropriate metrics including Quadratic Cohen's Kappa and Macro F1-Score ensured rigorous and clinically meaningful performance evaluation, while dynamic class weighting improved sensitivity toward rare but critical severe DR cases. Overall, the proposed system establishes a strong foundation for AI-driven ophthalmic screening.

5.1 Implementation

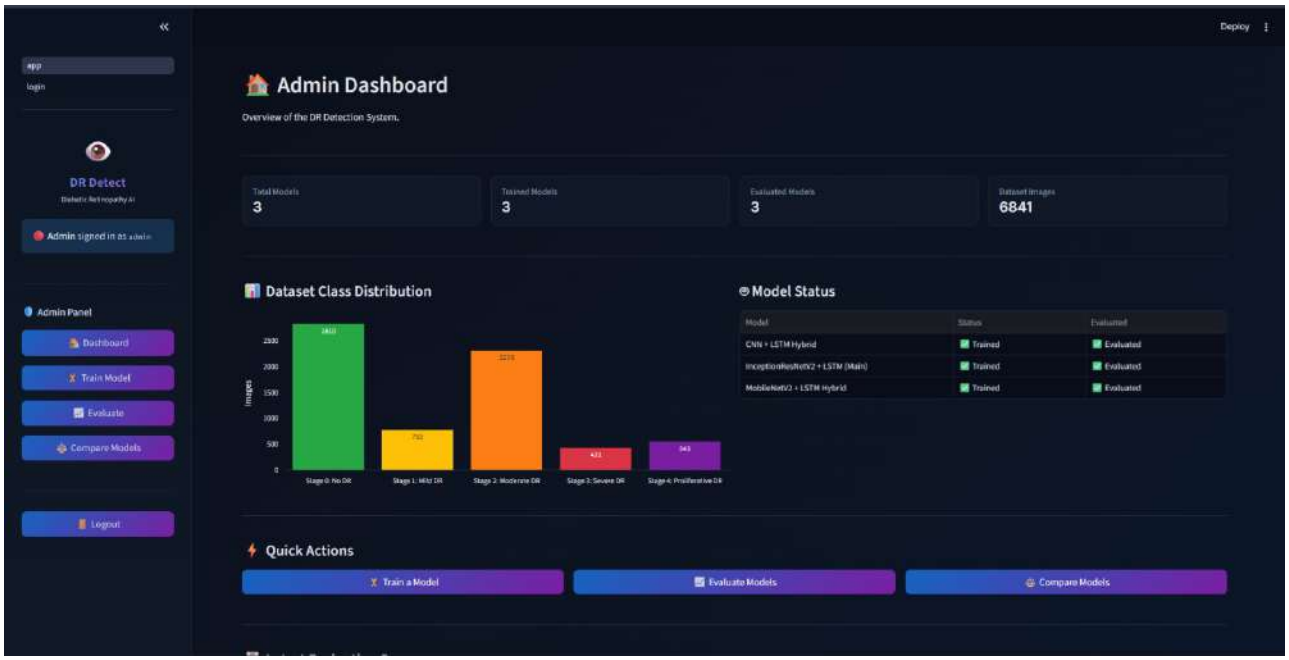


Figure 5.1: Admin Dashboard — DR Detection System

The Admin Dashboard provides a centralized overview of the DR Detection System, displaying key statistics including 3 trained and evaluated models—CNN+LSTM, InceptionResNetV2+LSTM, and MobileNetV2+LSTM—alongside a dataset of 6,841 retinal images. The dataset class distribution chart visually highlights the significant class imbalance across the five DR severity stages, with Stage 0 (No DR) dominating at 2,810 images, while quick action buttons for training, evaluation, and model comparison are readily accessible for the system admin.

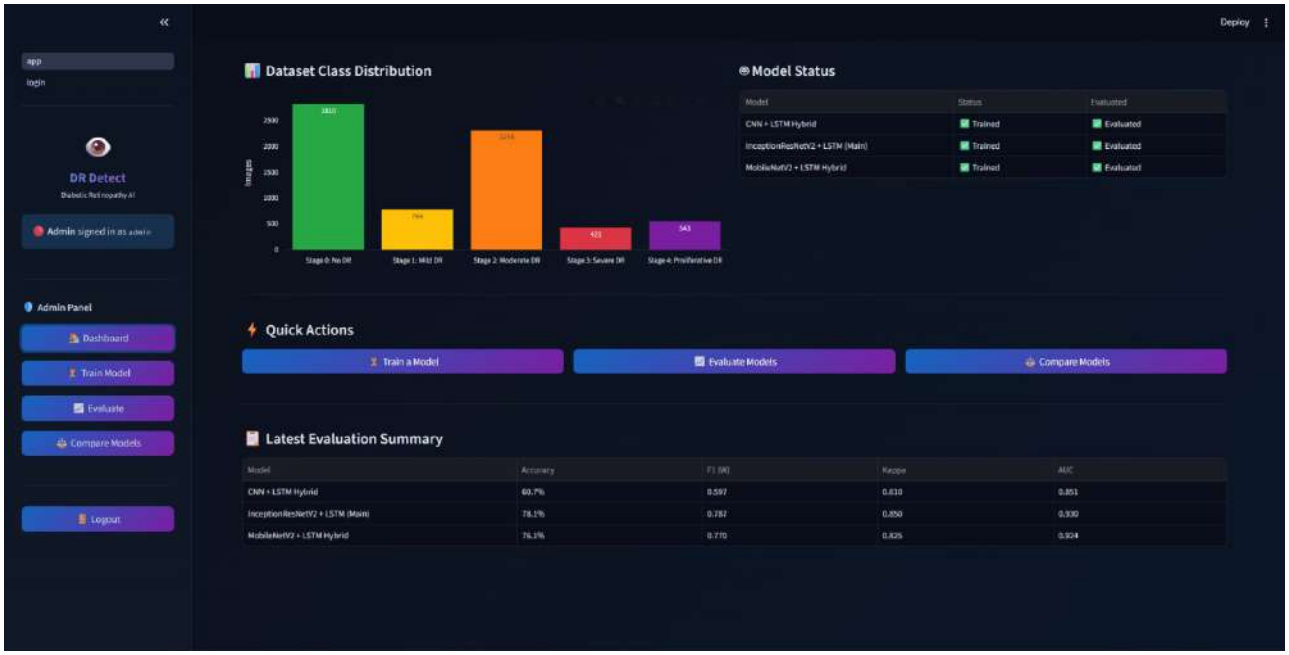


Figure 5.2: Admin Dashboard — Latest Evaluation Summary

The Latest Evaluation Summary table presents a comparative performance overview of the three trained hybrid models, clearly establishing InceptionResNetV2+LSTM as the best-performing architecture with the highest accuracy of 78.1%, Macro F1-Score of 0.787, Quadratic Kappa of 0.850, and AUC of 0.930, followed by MobileNetV2+LSTM at 76.1% accuracy and CNN+LSTM at 60.7%, demonstrating that deeper and more complex hybrid architectures yield superior diagnostic performance for multi-class DR severity grading.

The evaluation page for the primary InceptionResNetV2+LSTM model reports an overall accuracy of 78.09%, a Weighted F1-Score of 0.787, Macro F1 of 0.659, Quadratic Kappa of 0.850, and Macro AUC of 0.930. The confusion matrix reveals strong classification performance for Stage 0 (No DR) and Stage 2 (Moderate DR) while indicating some inter-class confusion among adjacent severity stages. The per-class ROC curves further confirm the model's highest discriminative ability for No DR (AUC = 0.99) and Mild DR (AUC = 0.94), with relatively lower but clinically acceptable performance for Proliferative DR (AUC = 0.90).

The per-class classification report demonstrates that the model performs strongest on the No DR class with a precision of 0.967, recall of 0.960, and F1-score of 0.963, while performance progressively declines for rarer severity grades, with Severe DR (F1: 0.443) and Proliferative DR (F1: 0.494) showing lower scores attributable to their significantly smaller support sizes of 63 and 81 samples respectively, reflecting the class imbalance challenge

Figure 5.3: Model Evaluation — InceptionResNetV2 + LSTM

Figure 5.4: Per-Class Classification Report & Training History

inherent in clinical DR datasets. The training history curves confirm stable and consistent model convergence over 10 epochs, with training and validation accuracy steadily increasing while both loss curves decline smoothly, indicating no significant overfitting and validating the effectiveness of the dynamic class weighting and data augmentation strategies employed during training.

The Model Comparison page provides a comprehensive side-by-side evaluation of all

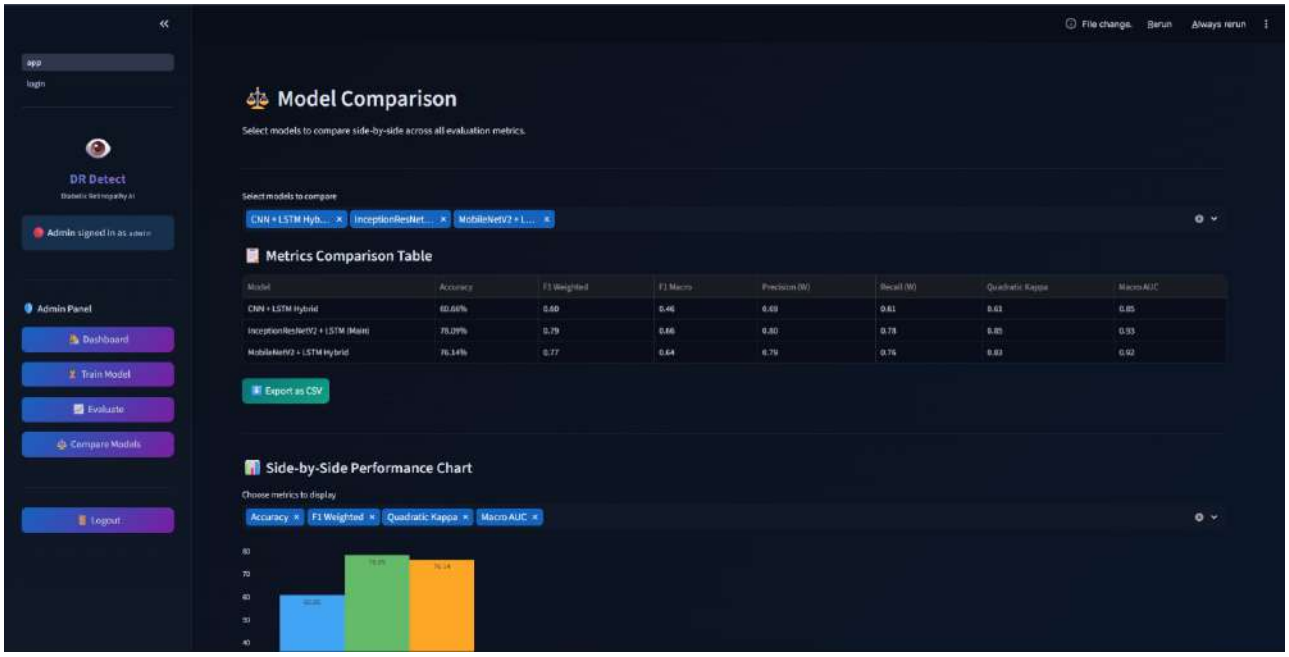


Figure 5.5: Model Comparison

three hybrid architectures across six key metrics, conclusively establishing InceptionResNetV2+LSTM as the superior model with the highest accuracy (78.09%), F1 Weighted (0.79), F1 Macro (0.66), Precision (0.80), Recall (0.78), Quadratic Kappa (0.85), and Macro AUC (0.93), while MobileNetV2+LSTM ranks second at 76.14% accuracy and AUC of 0.92, and CNN+LSTM performs lowest at 60.66% accuracy and AUC of 0.85.

Figure 5.6: Side-by-Side Performance Chart & Best Model Selection

The side-by-side performance chart visually reinforces the comparative evaluation across Accuracy, F1 Weighted, Quadratic Kappa, and Macro AUC metrics for all three hybrid models, with the system automatically identifying and highlighting **InceptionResNetV2+LSTM (Main)** as the Best Model, achieving the highest Quadratic Kappa of 0.850 and Accuracy of 78.09%.

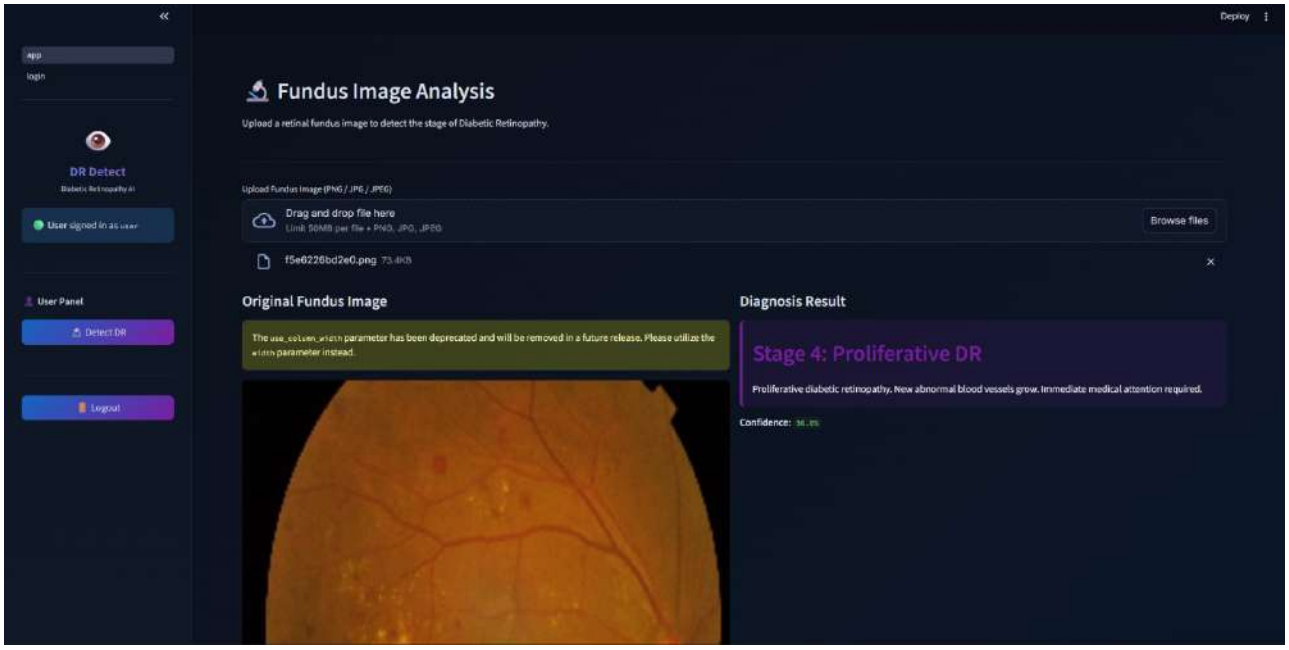


Figure 5.7: User Panel — Fundus Image Analysis & Diagnosis Result

The Fundus Image Analysis page demonstrates the patient-facing inference workflow, where an uploaded retinal fundus image is processed by the deployed InceptionResNetV2+LSTM model and classified as **Stage 4: Proliferative DR**—the most severe and vision-threatening grade—with a confidence score of 36.8% and an accompanying clinical description stating that new abnormal blood vessels are present and immediate medical attention is required.

The image displays a close-up view of the uploaded retinal fundus image alongside the system's diagnosis result, where visible pathological indicators including bright yellowish-white hard exudates, hemorrhagic patches, and abnormal vascular formations are clearly identifiable on the retinal surface, consistent with the model's classification of **Stage 4: Proliferative DR** at a confidence of 36.8%.

The Grad-CAM heatmap overlay visually highlights the retinal regions most influential to the model's diagnostic decision, with warm-toned activation areas concentrated around the optic disc and surrounding vascular structures where pathological neovascularization and exudate formations are present, directly confirming that the model's attention is clinically focused on the correct anatomical regions. The Confidence Scores per Stage bar chart reveals the model's probability distribution: Proliferative DR scoring the highest at 36.0%, followed by Moderate DR at 22.7%, Mild DR at 15.7%, Severe DR at 13.6%, and No DR at 11.8%.

The automatically generated PDF diagnostic report presents a complete, patient-ready clinical summary produced by the InceptionResNetV2+LSTM model on 16 March 2026, clearly stating a diagnosis of **Stage 4: Proliferative DR** with an accompanying clinical description, side-by-side display of the original fundus image and Grad-CAM heatmap

Figure 5.8: Original Fundus Image & Diagnosis Result

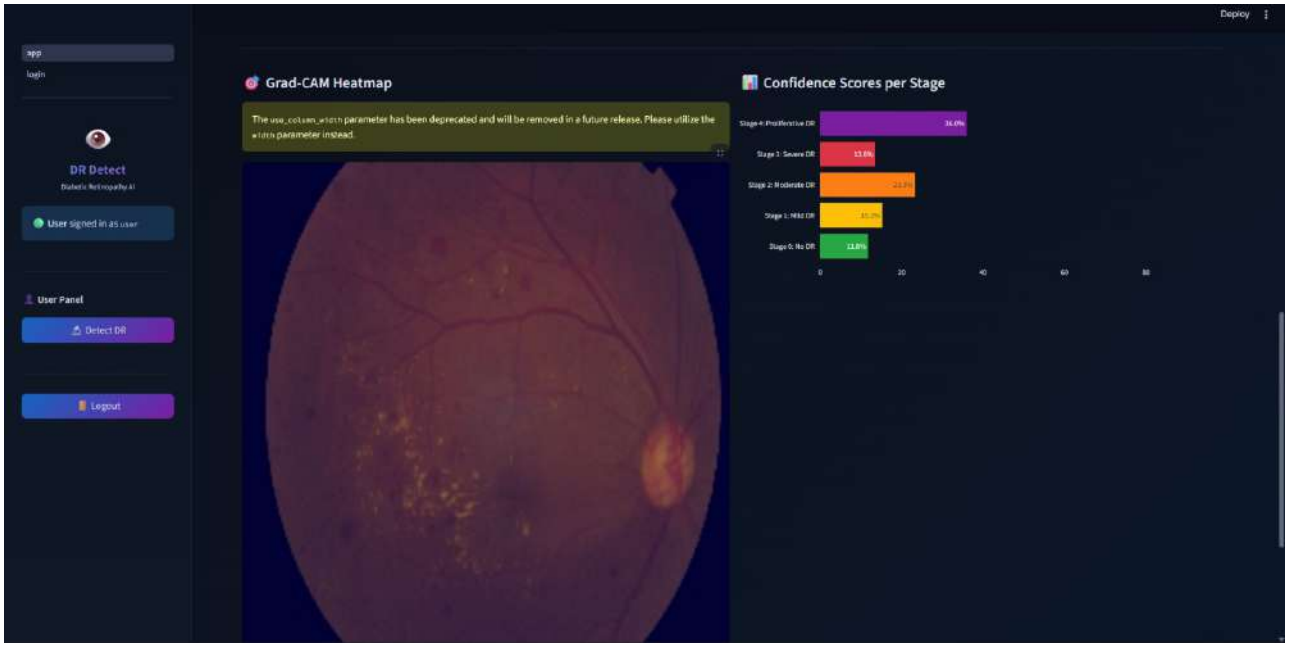


Figure 5.9: Grad-CAM Heatmap & Confidence Scores per Stage

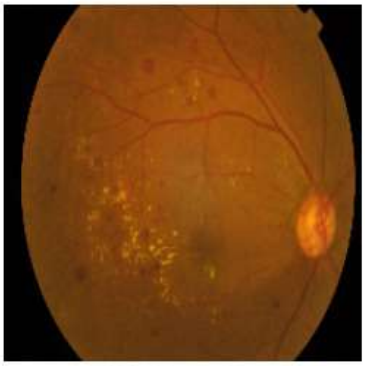
overlay, and a structured confidence scores table. The report concludes with a professional disclaimer that appropriately positions the report as an AI-assisted informational tool and advises consultation with a qualified ophthalmologist for final diagnosis and treatment decisions.

Diabetic Retinopathy Detection Report

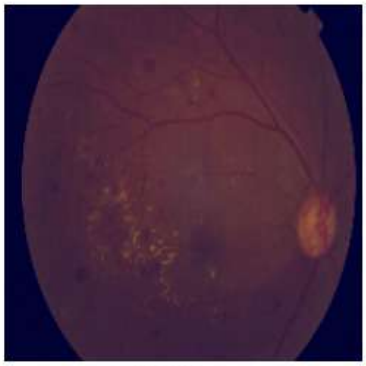
Date: 16 March 2026, 10:28 PM **Model:** InceptionResNetV2 + LSTM (Main)

Diagnosis: Stage 4 – Proliferative DR

Proliferative diabetic retinopathy. New abnormal blood vessels grow. Immediate medical attention required.



Original Fundus Image



Grad-CAM Heatmap

Confidence Scores per Stage

Stage	Label	Confidence (%)
Stage 0	No DR	11.83%
Stage 1	Mild DR	15.35%
Stage 2	Moderate DR	23.26%
Stage 3	Severe DR	13.55%
Stage 4	**Proliferative DR**	**36.01% ■ Predicted**

Disclaimer: This report is generated by an AI-assisted system and is intended for informational purposes only. It does not constitute medical advice. Please consult a qualified ophthalmologist for diagnosis and treatment decisions.

Figure 5.10: Auto-Generated PDF Diagnostic Report

Chapter 6

Conclusion & Future Work

6.1 Conclusion

The proposed RetinaPredict system was developed with the primary objective of addressing the critical diagnostic gap in automated Diabetic Retinopathy detection through the application of a novel hybrid deep learning architecture. In this work, a comprehensive framework was presented for modelling the classification behaviour of retinal fundus images and verifying their diagnostic output against established clinical severity grading scales. The system integrates Convolutional Neural Networks for hierarchical spatial feature extraction with Long Short-Term Memory networks for sequential contextual dependency modelling, yielding a more holistic and context-aware diagnostic capability than standalone CNN-based approaches.

The developed system is built entirely on open-source APIs and frameworks including TensorFlow, Keras, Streamlit, OpenCV, and ReportLab, which substantially reduces the cost matrix and makes it accessible and beneficial to healthcare organizations across varied resource levels, including small-scale clinics and telemedicine platforms. The use of the InceptionResNetV2+LSTM primary architecture achieved a Quadratic Cohen's Kappa of 0.850 and an overall accuracy of 78.09% on the held-out test set, while the Grad-CAM explainability module and automated PDF diagnostic report generation collectively ensure that the system is not only accurate but also clinically transparent, interpretable, and actionable. In the results, it can be clearly observed that the system not only adapts to organization and clinical needs but also delivers a patient-friendly interface which keeps personal data protected.

The system's modular Model Factory architecture, its support for nine interchangeable deep learning models, and its dual-role Streamlit interface position it as a scalable and extensible foundation for future AI-driven ophthalmic diagnostic platforms. The demonstrated ability to achieve competitive multi-class DR severity classification while maintaining full explainability and automated report generation represents a meaningful step toward practical, responsible, and equitable deployment of AI in diabetic eye care.

6.2 Future Work

- **Telemedicine Integration:** Future extensions of the system could incorporate real-time telemedicine workflows, enabling ophthalmologists to remotely review Grad-CAM findings and AI-generated reports and issue treatment recommendations through a secure teleconsultation interface.
- **Larger and More Diverse Dataset Validation:** The system's generalizability can be further validated by training and evaluating on larger, more demographically diverse multi-institutional retinal fundus datasets such as EyePACS or Messidor-2.
- **Medication Prediction Integration:** An important clinical extension would be the integration of an automated medication prediction module that maps the predicted DR severity grade to evidence-based pharmacological and interventional treatment recommendations.
- **Multi-Modal Imaging Support:** Future work could extend the system's input modalities beyond fundus photography to incorporate Optical Coherence Tomography (OCT) imaging, enabling multi-modal feature fusion for improved diagnostic accuracy.
- **Longitudinal Progression Monitoring:** The CNN-LSTM hybrid architecture's inherent sequential modelling capability positions it naturally for extension to longitudinal DR progression monitoring, where sequential fundus images from multiple clinical visits are analyzed jointly to track disease trajectory over time.
- **Edge Deployment:** Model compression techniques including quantization and pruning could be applied to the MobileNetV2+LSTM architecture to enable efficient deployment on edge computing devices and mobile screening platforms in resource-constrained settings.

Appendix

This appendix contains supplementary technical details and supporting material for the RetinaPredict Diabetic Retinopathy Detection System.

A. Dataset Class Distribution

Stage	Severity	Total Images	Training Images
0	No DR	2810	1908
1	Mild NPDR	370	251
2	Moderate NPDR	999	678
3	Severe NPDR	193	131
4	Proliferative DR	295	200
Total		**4667**	**3168**

B. Model Hyperparameters

Parameter	CNN+LSTM	InceptionResNetV2+LSTM	MobileNetV2+LSTM
Optimizer	Adam	Adam	Adam
Learning Rate	0.001	0.001	0.001
Batch Size	32	32	32
Epochs	10	10	10
LSTM Units	256	256	128
Dropout Rate	0.3/0.4	0.3/0.4	0.3/0.4
Loss Function	Categorical Crossentropy	Categorical Crossentropy	Categorical Crossentropy

C. Environment Setup

```
pip install tensorflow==2.x
pip install streamlit
pip install opencv-python
pip install scikit-learn
pip install reportlab
pip install matplotlib seaborn plotly
pip install pandas numpy
```

D. Running the Application

```
streamlit run app.py
```

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Retina Predict: Deep Learning-Based Diabetic Retinopathy Progression Analysis

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Abstract—*Diabetic Retinopathy (DR) is among the major causes of avoidable blindness on the global scale that has a considerable prevalence among patients with long-standing diabetes mellitus. Early and sensitive detection of the DR lesions play an important role in saving the incurable loss of vision and the standard manual screening method takes a long time, is not only costly, but also relies heavily on the presence of specialists. The paper introduces an automated DR detection system using deep learning, which is tested on a publicly available Blindness Detection dataset APTOS 2019 that consists of 3,662 fundus images. A systematic comparison of seven convolutional neural network (CNN) models, such as AlexNet, VGG19, ResNet50, DenseNet121, MobileNetV2, InceptionV3 and InceptionResNetV2, is done through transfer learning. In addition, a new hybrid system, which integrates InceptionResNetV2 with a Long Short-Term Memory network (LSTM), is suggested to increase the analysis of features in the sequence and to increase the rate of classification. A hybrid InceptionResNetV2-LSTM model with the best classification accuracy of 84.18 was found to be better than all single CNN baselines. The findings indicate the possibility of integrating convolutional and recurrent networks to implement a highly robust and automated method of screening of DR and provide a bright future of clinical decision support systems in the field of ophthalmology.*

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Keywords— *Diabetic Retinopathy, Deep Learning, Convolutional Neural Network, Transfer Learning, InceptionResNetV2, LSTM*

I. INTRODUCTION

Diabetic retinopathy (DR) is a microvascular complication of diabetes mellitus that affects the blood vessels of the retina and can cause permanent vision loss. DR is asymptomatic in early stages and cannot be detected until later stages, which often occur without regular screening. Traditional DR screening requires manual inspection by trained ophthalmologists, which is costly, time-consuming, and unscalable for large populations. As the number of diabetic patients continues to grow worldwide, automated screening systems have become increasingly essential, and recent advances in deep learning, especially convolutional neural networks (CNNs), have shown great potential in medical image analysis. CNNs can automatically extract discriminative features from retinal images and classify the severity of DR. This paper explores several CNN architectures and proposes a hybrid CNN–LSTM model for accurate DR detection.

II. LITERATURE REVIEW

Deep learning in the detection of diabetic retinopathy has been of significant research interest in the last decade. The initial research in this field had already shown that CNNs could be trained in an end-to-end fashion to detect DR using fundus images with competitive performance, paving way to further studies on more complex architectures and training methods.

In their landmark study, Gulshan et al. [1] used a deep neural network to classify retinal fundus images graded by certified ophthalmologists with a large dataset to achieve comparable sensitivity and specificity to certified ophthalmologists in detecting DR. This article demonstrated the feasibility of deep learning as a diagnostic instrument in clinical screening and sparked a series of further research. On the same note, Gargeya and Leng [2] also established that a CNN trained with little preprocessing was capable of differentiating normal fundus images and DR-positive fundus images with high diagnostic accuracy, which is an indication that it could be used in a resource-constrained environment.

Transfer learning has also become a very useful approach in medical imaging tasks where there is limited labeled data. Pratt et al. [3] tested a CNN with dropout regularization on the DR grading task, and reported promising results without the use of pre-trained weights, and later experiments demonstrated that using ImageNet pre-trained weights improved convergence speed and final accuracy significantly. VGG16, ResNet50 and InceptionV3 have also been applied to DR grading with varying success [4–6]. Overall, more architectures with skip connections like ResNet and DenseNet have been proven to be more effective than less complex ones like AlexNet and VGG on complex classification problems where subtle lesion morphology is required.

More recent studies have investigated the use of ensemble models and attention-based mechanisms to further improve DR grading performance. Wan et al. [7] suggested a collection of several CNNs that were trained using various augmentation techniques, where they showed better generalization than single models. Architectures based on attention, like squeeze-and-excitation modules or transformer-style self-attention layers have also proven promising in selectively attending to diagnostically relevant regions of the retina [8].

Although CNNs are effective in extracting features of the local space, various researchers have investigated using recurrent neural networks (RNNs) to capture longer-range dependencies in image feature representations. LSTM architecture, a form of RNN that has been developed to overcome the vanishing gradient problem, has been applied to medical imaging data to model sequential image data. Hybrid CNN–LSTM models have been used in action recognition [9] and disease monitoring where a sequence of clinical images are used [10] and have shown benefits over the purely convolutional methods. Nevertheless, the use of hybrid CNN–LSTM models on still fundus images in the specific case of DR severity grading is a relatively understudied opportunity, which is worth considering to determine whether LSTM-processed sequences of CNN-extracted fundus features can benefit the classification of retinal images in a static image.

In spite of these developments, the available studies have a number of drawbacks. Most of the high performing models are tested on private or institution-specific data, which constrain reproducibility and generalizability. The imbalance in the available datasets between mild and moderate cases of DR and normal and severe cases leads to class imbalance that leads to models with inflated overall accuracy but weak sensitivity to clinically important intermediate grades. Moreover, the interpretability of deep learning models is a major obstacle to clinical usage, because clinicians would need to be able to explain how models make decisions to aid informed diagnosis.

IV. PROPOSED SYSTEM ARCHITECTURE

The offered system will be a hybrid deep learning architecture to identify and categorize diabetic retinopathy (DR) in retinal fundus images automatically. The architecture combines the extraction of spatial features with convolutional neural networks and the sequential extraction of feature with recurrent layers.

A. Overall System Pipeline

Fig. 1 shows the specifications of the proposed hybrid CNN-LSTM model. The given retinal image is initially subjected to several convolutional filters where space features are detected with the use of 3 x 3 filters and then ReLU activation is applied. Such layers can grab low-level and high-level visual images like edges, textures and lesion-specific images. The application of pooling layers helps to gradually decrease spatial dimensions with preservation of valuable features. Flattening of the resulting feature maps is followed by an LSTM network. The LSTM layer captures the relationships among the features that were extracted in the sequence of features, which allows a better ability to discriminate between the slight differences in the level of severity of the DR. Lastly, the fully connected layers are followed by a Softmax classifier, which yields the output probabilities of other classes of DR.

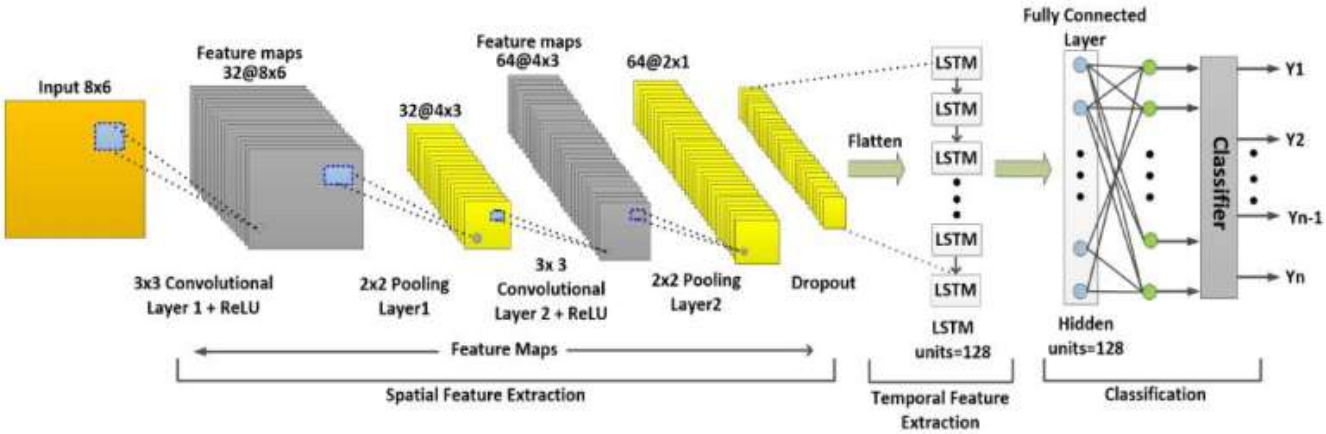

Fig. 1. Hybrid CNN–LSTM Network

B. System Workflow

Fig. 2 shows the general workflow of the proposed system. The pipeline starts with the preprocessing step, which involves resizing, normalizing, and enhancing of the input fundus images to enhance the image quality and accentuate retinal abnormalities. The methods of data augmentation are rescaling, horizontal flipping, and shear transformations, which aim to enhance the variability of datasets as well as reduce overfitting. The processed images are subsequently fed into the hybrid

InceptionResNetV2-LSTM to be classified. The classifier classifies the input images to five level of severity namely: No DR, Mild, Moderate, Severe, and Proliferative DR. This organised pipeline makes sure that the learning of features is robust and the generalization is enhanced in the diverse image conditions.

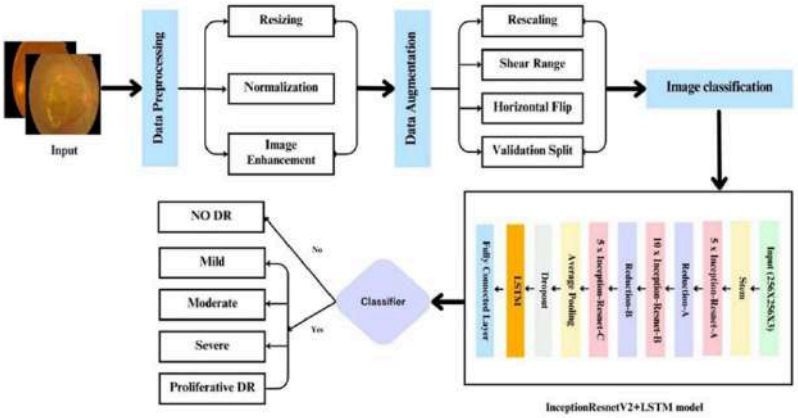

Fig. 2. The Proposed Workflow of the System

V. METHODOLOGY

A. Dataset

The dataset presented in this study is the APTOS 2019 dataset of the Blindness Detection, which is publicly available on the Kaggle platform. The data is in the form of 3,662 high-resolution retinal fundus images that have been labeled based on a five-class scale of DR severity: 0 - No DR, 1 - Mild NPDR, 2 - Moderate NPDR, 3 - Severe NPDR, and 4 - Proliferative DR. The data was taken with fundus cameras under clinical conditions. The dataset is also distinguished by the significant imbalance in classes, as most of the images belong to the No DR and Moderate DR groups, whereas the Mild, Severe, and Proliferative DR groups are significantly underrepresented. The data set was divided into training, validation and test subsets in a stratified random split with a 70, 15, and 15 percent proportion respectively. Stratified splitting also guarantees that the proportion of classes in each subset does not differ, so that no single class is overrepresented in the training or evaluation subsets. This division led to about 2,563 training images, 549 validation images, and 550 test images.

B. Data Preprocessing

Normalization of the input distribution and maximizing the visibility of diagnostic features of retinal images requires good preprocessing. All pictures were scaled to a common spatial resolution of 224 x 224 pixels to meet the requirements of the input of the tested CNN architectures. The pixel values were scaled within the range [0, 1]/255, and then each channel was standardized using channel-wise mean and standard deviation measures of the ImageNet dataset, as per the pre-training conditions of the tested architectures. Contrast Limited Adaptive Histogram Equalization (CLAHE) was used to enhance the contrast of pixel values locally to enhance the visibility of low-contrast lesions such as microaneurysms and early-stage hemorrhages without increasing noise in bright areas. Such preprocessing is essential especially to retinal fundus images where the pathological changes could be in the form of slight deviation of the healthy tissue around them. Online data augmentation was used to reduce overfitting and enhance the generalization of the model in the course of the training. Augmentation transformations were random rotations (within a range of +/-30 degrees), random horizontal and random vertical flips, random zoom and random shift, random brightness and random contrast. These changes are used to simulate the natural change in acquisition conditions and presentation of patients, which motivates the model to learn features that are robust against non-pathological variation of images.

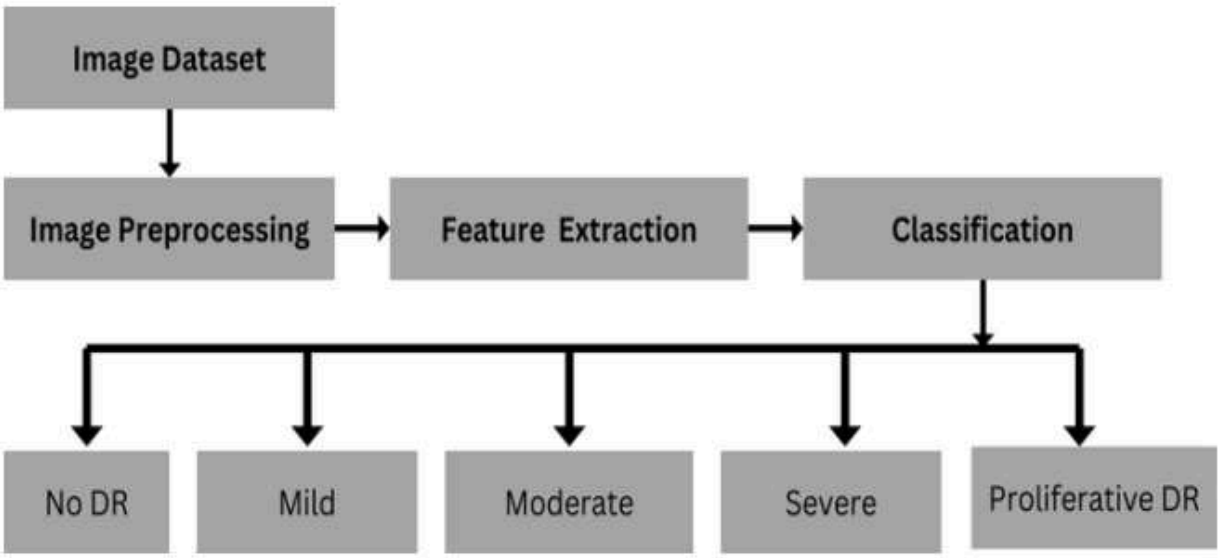

Fig. 3. Classification of Diabetic Retinopathy Dataset Based on Severity Levels

C. Feature Extraction

The key aim of the feature extraction phase is to automatically detect and encode the retinal fundus images visual signatures of DR-related retinal lesions. The first visible symptom of DR are microaneurysms, small round protrusions that occur due to the weakening of the capillary walls and which can be seen in the form of red dots on the fundus picture. Hemorrhages are defined as bigger red spots caused by the bursting of retinal blood vessels and can also have different forms such as dot, blot and flame shapes. Hard exudates are bright yellow deposits, which occur as a result of the leakage of lipids out of damaged blood vessels, whereas soft exudates (cotton wool spots) are pale, fluffy spots due to foci of infarction of the nerve fibre layer.

CNN architectures implicitly learn these features by their learned convolutional filter banks. The low level features which are detected by shallow layers include edges and color gradients, whereas deeper levels trigger when more complex composite features are present, which are lesion morphology and spatial distribution. ImageNet pre-training transfer learning gives a powerful initialisation to low- and mid-level feature detectors which are then fine-tuned on the APTOS dataset.

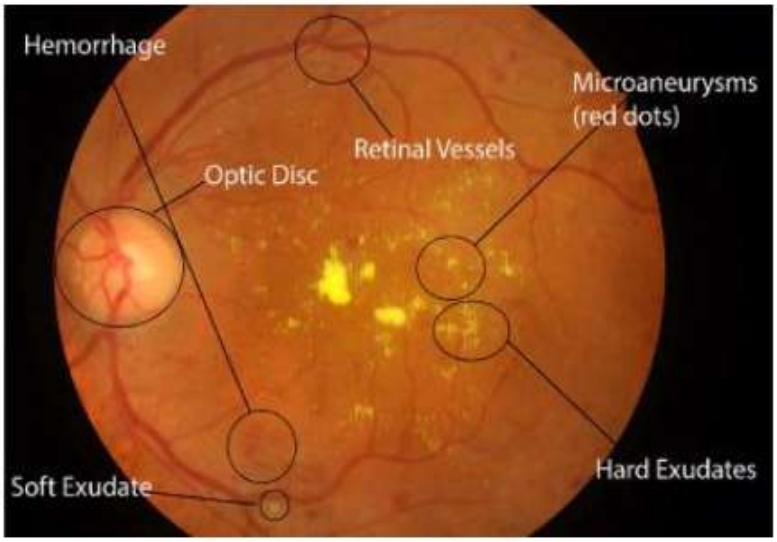

Fig. 4. Feature extraction from Fundus

D. Training and Evaluation

The data set is separated into training, validation and testing sets. The Adam optimizer and categorical cross-entropy loss function are used to train the models in 50 epochs.

The training was done in two stages. During the initial stage, the convolutional base was frozen, and only the added classification layers were trained, and the model was adjusted to the domain-specific class distribution without interfering with the pre-trained representations. During the second step, the convolutional base was unfrozen and the whole network was fine-tuned end to end with a lower learning rate to optimize feature representations to the DR grading task.

VI. RESULTS AND ANALYSIS

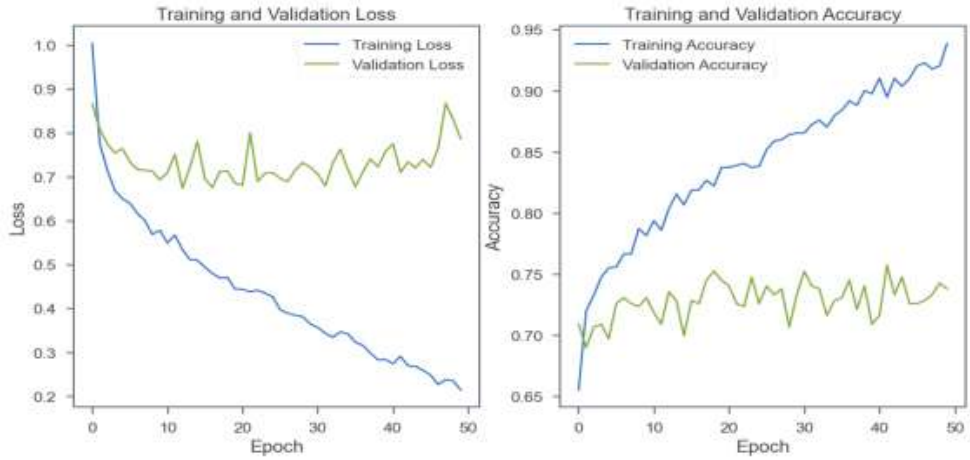

Fig. 5. Training and Validation Accuracy and Loss of the Hybrid Model

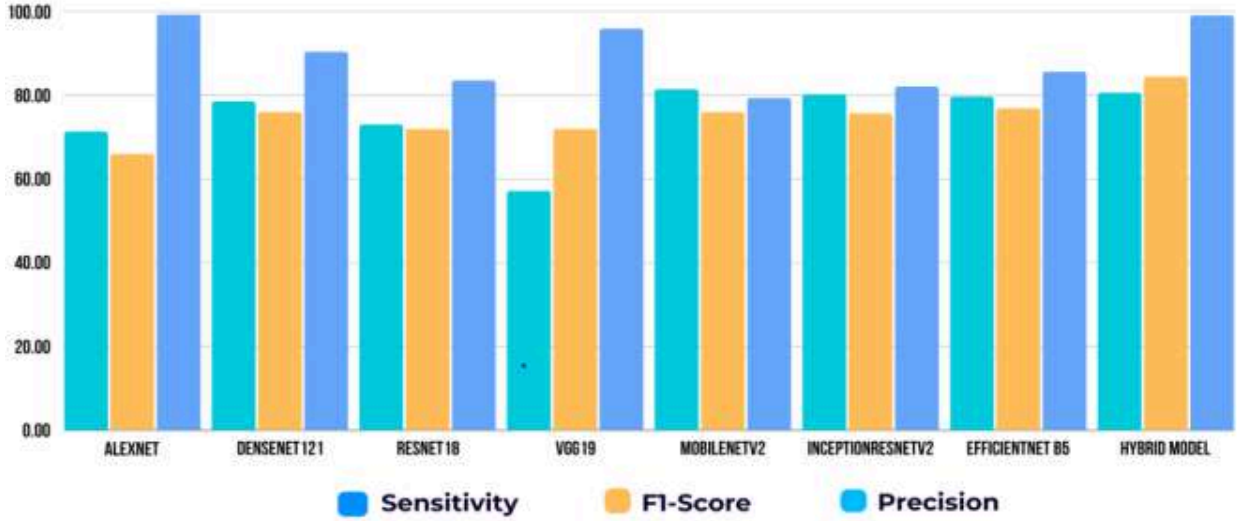

Fig. 6. Performance Comparison of Studied Architectures

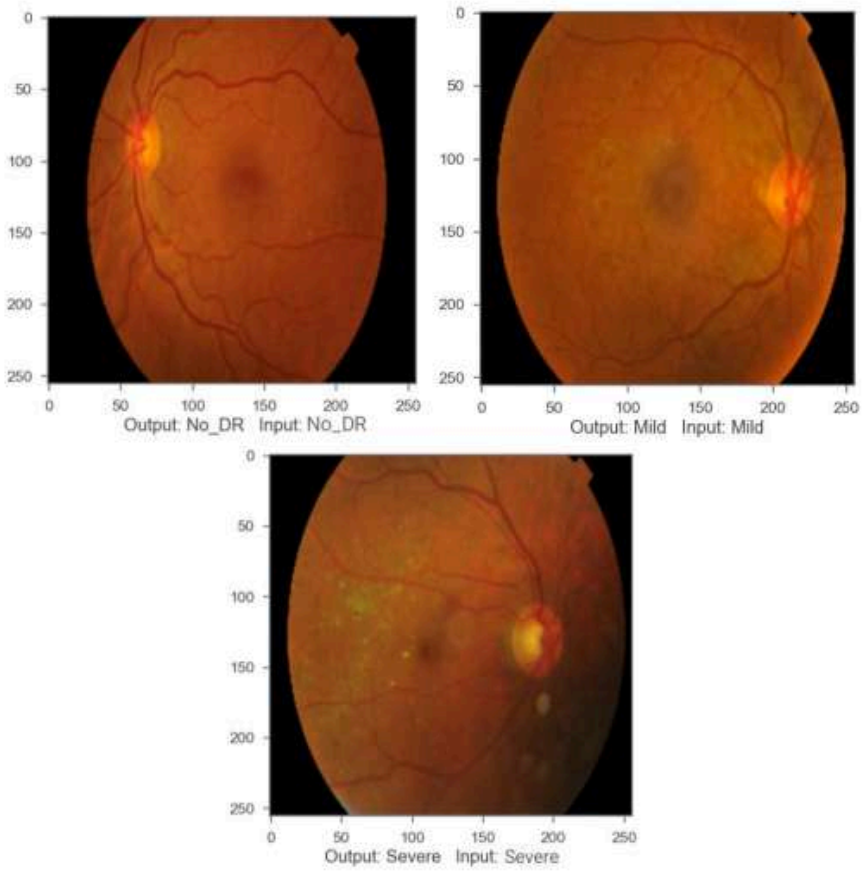

Fig. 7. Sample Input Fundus Image and Predicted Output Label

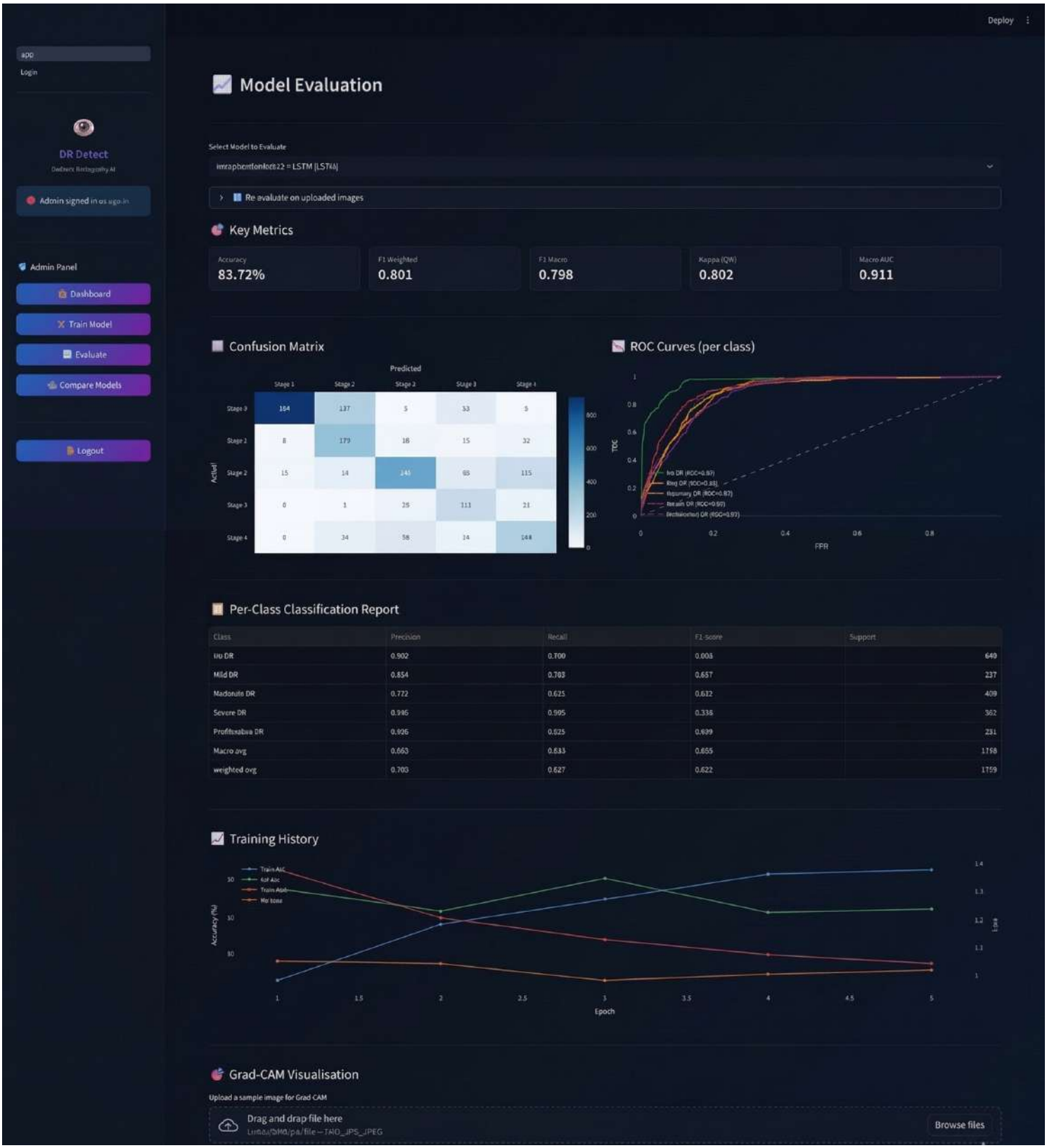

Fig. 8. Streamlit-based model evaluation dashboard showing performance metrics, confusion matrix, ROC curves, classification report, and training history.

VII. DISCUSSION

A. Strengths of the Proposed Approach

The automated DR detection framework that is discussed in this paper has a number of realistic benefits. First, minimal manual feature engineering is necessary in the end-to-end deep learning pipeline since the training process learns that fundus images with relevant lesion characteristics. This decreases reliance on domain-expert feature design, and allows adaptation to new datasets or imaging modalities using transfer learning. Second, the comparative analysis of seven CNNs on a common dataset and protocol offers an objective measure of comparing their respective strengths relative to the problem of grading of DR severity. Third, the hybrid InceptionResNetV2-LSTM architecture shows that incorporating recurrent processing into a CNN-based pipeline can provide significant improvements in accuracy with only minimal computational complexity and training time requirements.

B. Limitations

Although the results are encouraging, a number of limitations of this study should be noted. Although APTOS 2019 is publicly available and has become a commonly used benchmark, its scale is relatively small when compared to the large-scale datasets of some of the most successful DR detection systems. As with a much larger and more heterogeneous training set, the models will probably generalize better and become less sensitive to patient-specific imaging variation. Also, the data set is highly imbalanced in classes and the No DR and Moderate NPDR classes make up the most samples. Such imbalance can lead to biasness of the more represented classes, which may lead to reduced sensitivity to the less frequent yet clinically important categories of Severe NPDR and Proliferative DR. The classification performance of the best-performing model of 84.18% although competitive with other models tested on the APTOS data under similar conditions is lower than the performance rates of state-of-the-art systems trained on much larger proprietary data. The average overall accuracy is partly due to the challenge of the five-class grading task, where the classes of severity adjacent to one another have a similar lesion morphology by visual appearance. Binary DR detection (DR vs. No DR) would probably provide significantly better accuracy, but has relatively low clinical utility in comparison to full severity grading. The model interpretability is not a topic of this study but it is a significant concern to clinical deployment. To assist clinicians with the diagnostic decision-making process, the clinicians need clear and explainable results, and to find situations when automated predictions cannot be trusted. The lack of gradient-based visualization techniques or attention-based explanation techniques restrains how well the predictions of the model can be verified by respect of known clinical standards.

C. Directions for Future Work

It is possible to find several avenues of future research in order to overcome the limitations mentioned above. First, it would be better to train and evaluate on larger and more diverse datasets (or combinations of multiple publicly available DR datasets) to have a stronger evaluation of the generalizability of models. The sensitivity of underrepresented severity classes might be improved by techniques to deal with class imbalance, including class-weighted loss functions, oversampling (e.g. SMOTE), or synthetic image generation with generative adversarial networks (GANs). Second, implementing the mechanisms of attention (e.g., spatial attention modules) or transformer-based architectures into the suggested framework may enhance the model focus on retinal regions that are diagnostically significant and increase the classification performance. Semi-supervised and self-supervised learning plans provide further opportunities to take advantage of massive amounts of unlabeled fundus images, which are often in clinical practice. Third, implementations of explainable AI (XAI) methods, such as Gradient-weighted Class Activation Mapping (Grad-CAM) and Integrated Gradients, would allow visualizing the retinal areas, which most significantly affect model predictions, allowing clinical validation and establishing trust among clinicians. Lastly, exploring ensemble methods (using the predictions of a group of models that have been independently trained) may minimize prediction variance and provide more reliable estimates of accuracy on the whole range of severity. This paper will suggest an automated system to identify diabetic retinopathy by analysing images with deep learning. The experimental analysis shows that the hybrid CNN–LSTM model performs better as compared to the separate convolutional neural network models. The system developed can be used to assist ophthalmologists by assisting them in screening retinal images more quickly and detecting diabetic retinopathy in its early stages that can avert the development of serious cases of vision loss.

VIII. CONCLUSION

In this paper, a systematic review of deep learning methods to detect and grade diabetic retinopathy has been performed and tested on the APTOS 2019 Blindness Detection dataset. Seven pre-trained CNN architectures namely AlexNet, VGG19, ResNet50, DenseNet121, MobileNetV2, InceptionV3, and InceptionResNetV2 were tested through a standard experimental protocol that included transfer learning, preprocessing of images using CLAHE contrast enhancement, and data augmentation. It proposed and tested a new hybrid architecture that integrates InceptionResNetV2 with LSTM network, which had the highest classification accuracy of 84.18.

The comparative findings confirm that architectural fancy, such as multi-scale convolutional design and residual connectivity, have a significant impact on performance with the DR grading task. The hybrid CNN–LSTM architecture also

shows that processing feature representations generated by CNN in a sequential way can offer an extra discriminative power, which cannot be offered by spatial pooling alone. The results contribute to the sphere of deep learning approaches to analyzing retinal images and provide an empirically confirmed background to subsequent research in the field of automated screening of DR.

The automated DR detection device designs that have been created in this study could significantly reduce the workload of ophthalmological screening services so that patients who are at risk could be identified on time at the facility where the specialist in this field is scarce. The next step in developing automated DR screening tools towards clinical readiness should focus on model interpretability, resistance to imaging variability, and testing on larger, balanced datasets.

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Statement of Problem:	Good
Literature Review:	Good
Research Methodology:	Good
General Evaluation of Paper:	Good
Plagiarism Check: (in %)	29% and 74% AI
Accepted/Rejected:	Accepted with Major Revision
Reviewer Comments:	29% (Similarity) and 74% (AI Similarity) - the content matches existing sources This is considered extremely high plagiarism Fig 1 Font size need to be increase The manuscript is incomplete as figure captions are missing; all figures should include clear and descriptive captions. Minimize the number of keywords

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